

Pre-processing of EEG Signals

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HERBioLab

Pre-processing of EEG signals



Enhancing raw EEG data requires pre-processing; improving SNR is done by removing noise sources and artifacts.



Sweat, movement, heartbeat, and muscular activity contaminate EEG signals, as do power-line noise and improper electrode contact.

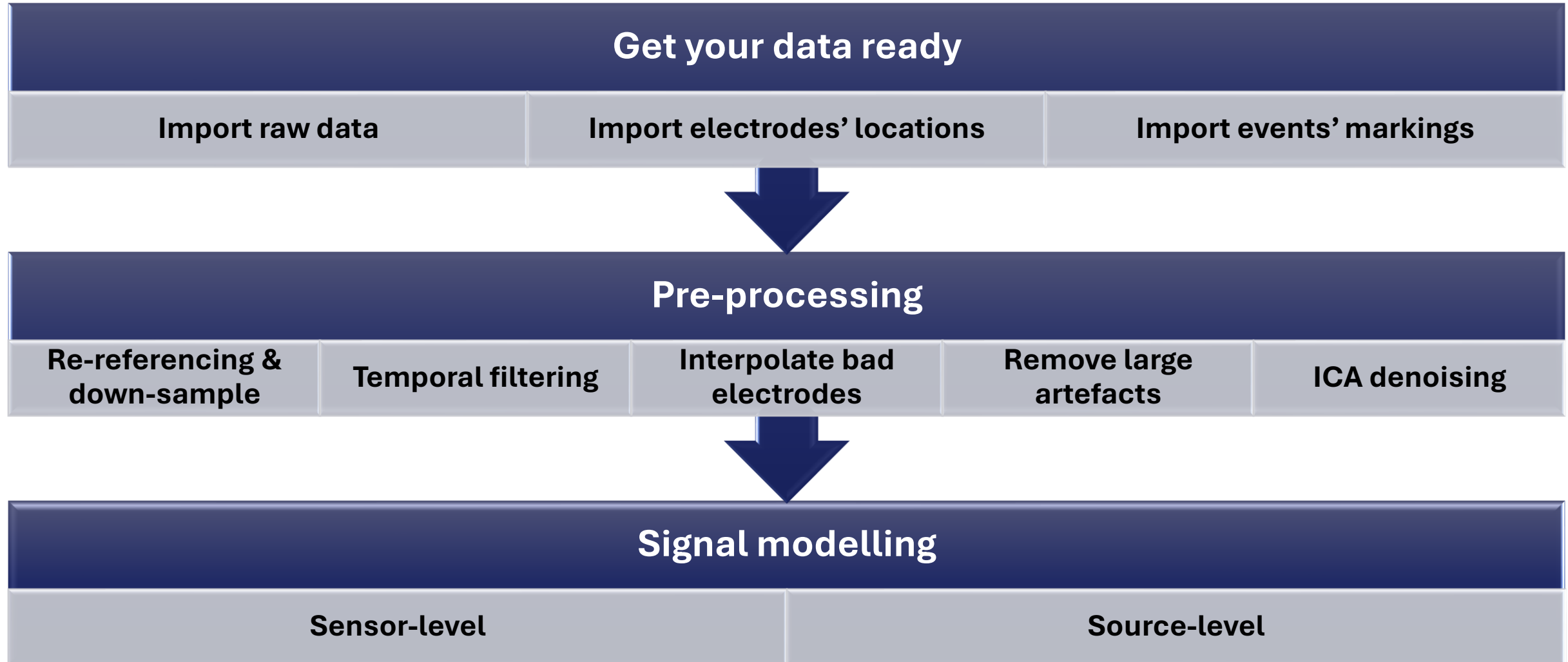


Manually examining data is most trustworthy, but time-consuming.



Automatic noise removal algorithms are extensively used to analyse larger datasets **but should be validated visually.**

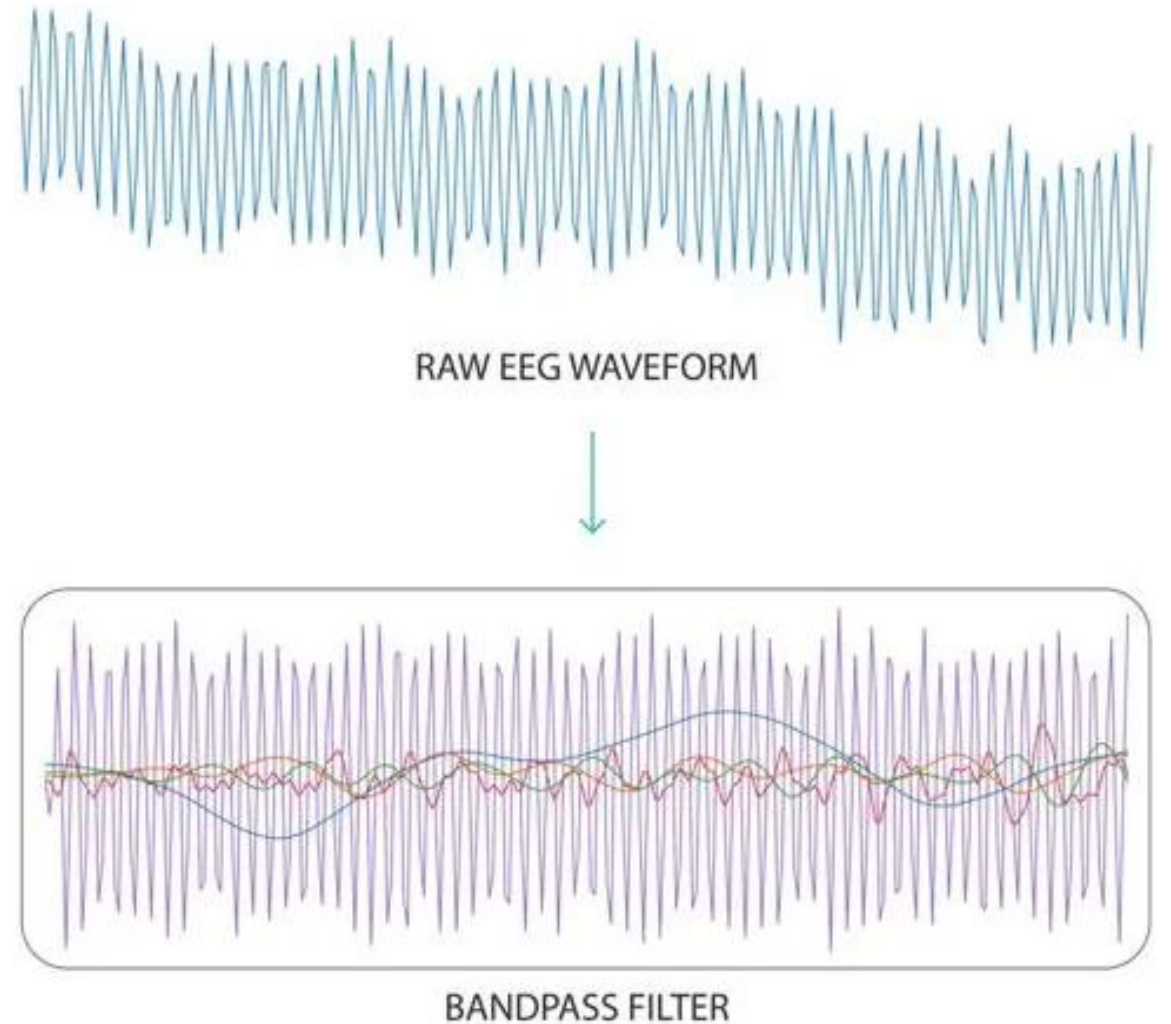
Let's create an EEG pipeline



Preprocessing – Temporal filtering

A temporal filter will remove all frequencies not related to the signal of interest.

Notch filters, especially for 50 Hz or 60 Hz, can be used to remove power-line-noise frequencies.

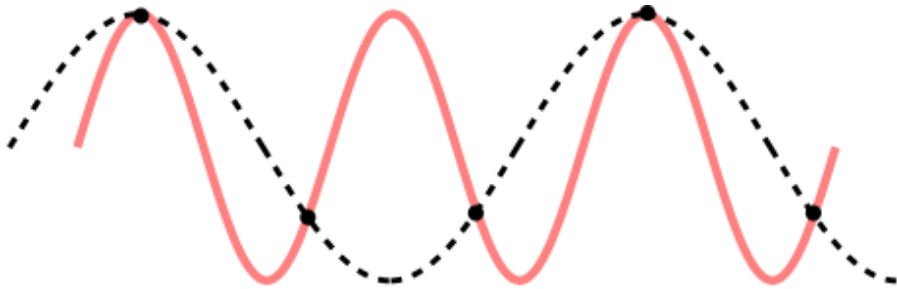


Preprocessing – Re-referencing and down-sampling

Down-sampling

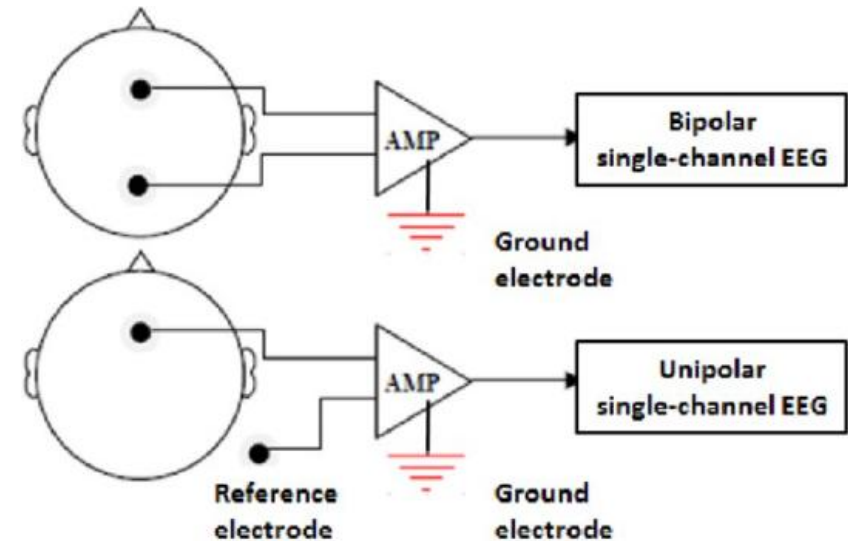
After temporal filtering, the EEG can be down-sampled.

- This will reduce the size of the data and speed up the processing.
- The down-sampling frequency should be **at least double** the highest remaining frequency in the data.



Re-referencing

Re-referencing is needed to reject the common-mode signal in post-processing.



Preprocessing – Interpolating bad channels

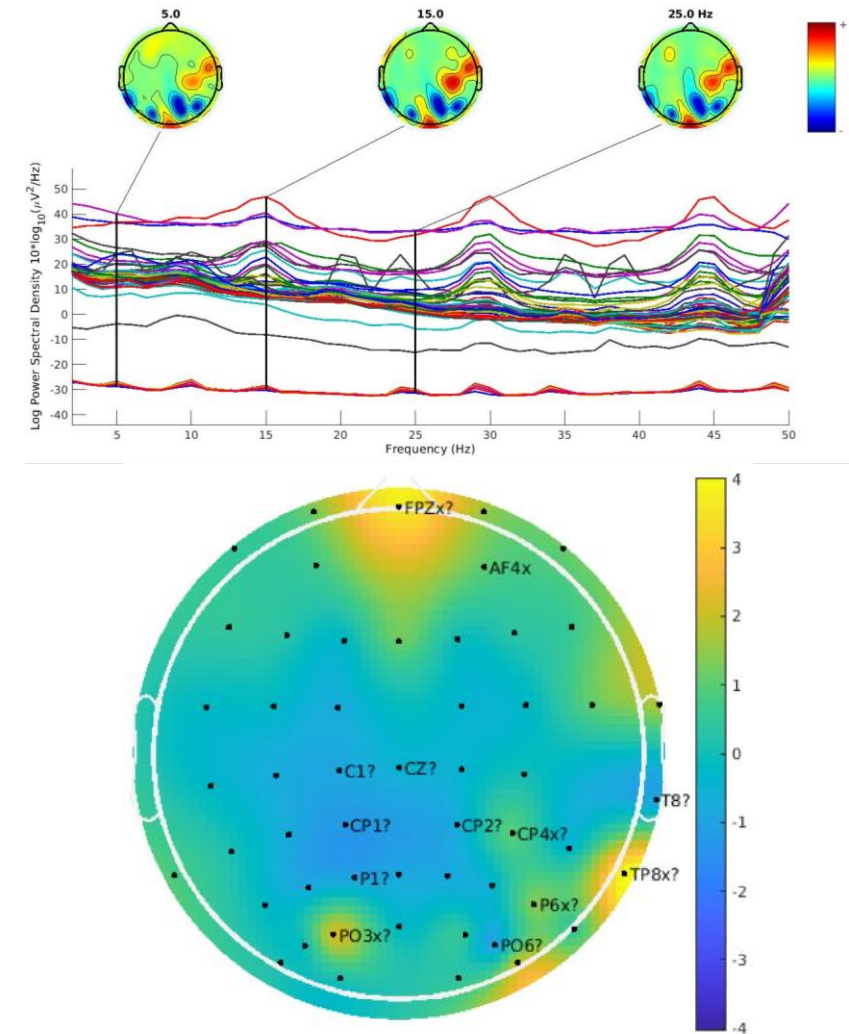
Due to displacement while recording or high impedances, some electrodes are best removed because of data corruption.

These electrodes can be interpolated in later processing using nearby electrodes.

Through visualizing the spectral maps, bad electrodes had out-of-EEG voltage range and/or smooth spectra.

Example of frequency spectra of EEG electrodes showing some smooth curves around -27 dB representing bad electrodes.

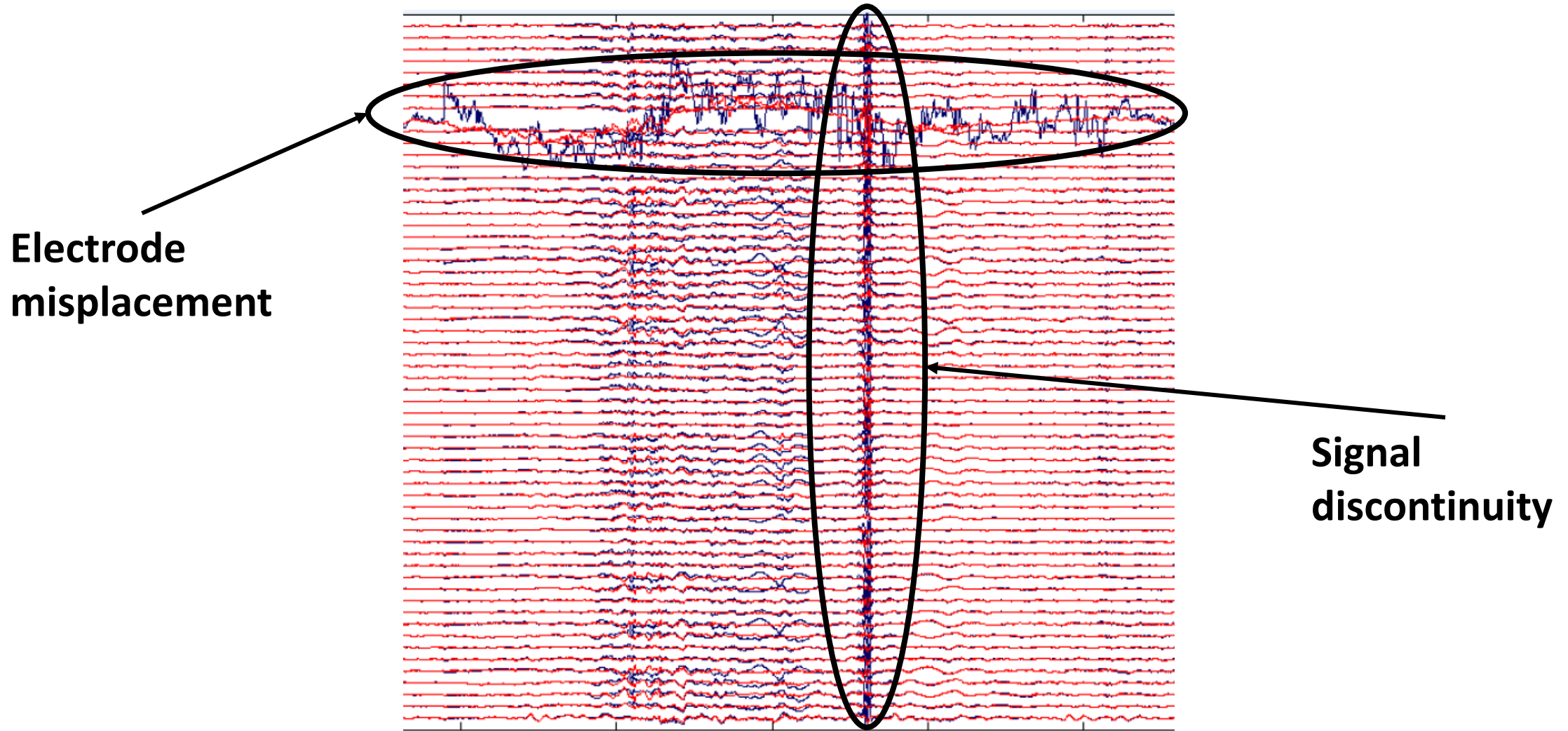
The number of electrodes interpolated was $\geq 25\%$ (i.e. ≥ 15 out of 60) or interpolated electrodes were adjacent to each other.



Different types of EEG artifacts and their origins

Physiological/internal				Extra-physiological/external		
Ocular	Cardiac	Muscle	Others	Instrumental	Interference	Movement
- Eye blink - Eye movement - REM sleep	- ECG pulse	- Swallowing - Talking - Scalp contraction - Chewing	- Sweat - Respiration	- Electrode displacement and pop-up - Cable movement - Poor ground	- Electrical - Magnetic - Sound - Optical	- Head movement - Body movement - Limbs movement - Other movements

Visualizing – Large EEG artefacts



Preprocessing of Large EEG artefacts – ASR

Artifact subspace reconstruction (ASR) is a non-stationary method dedicated to automatically detecting and removing artifacts.

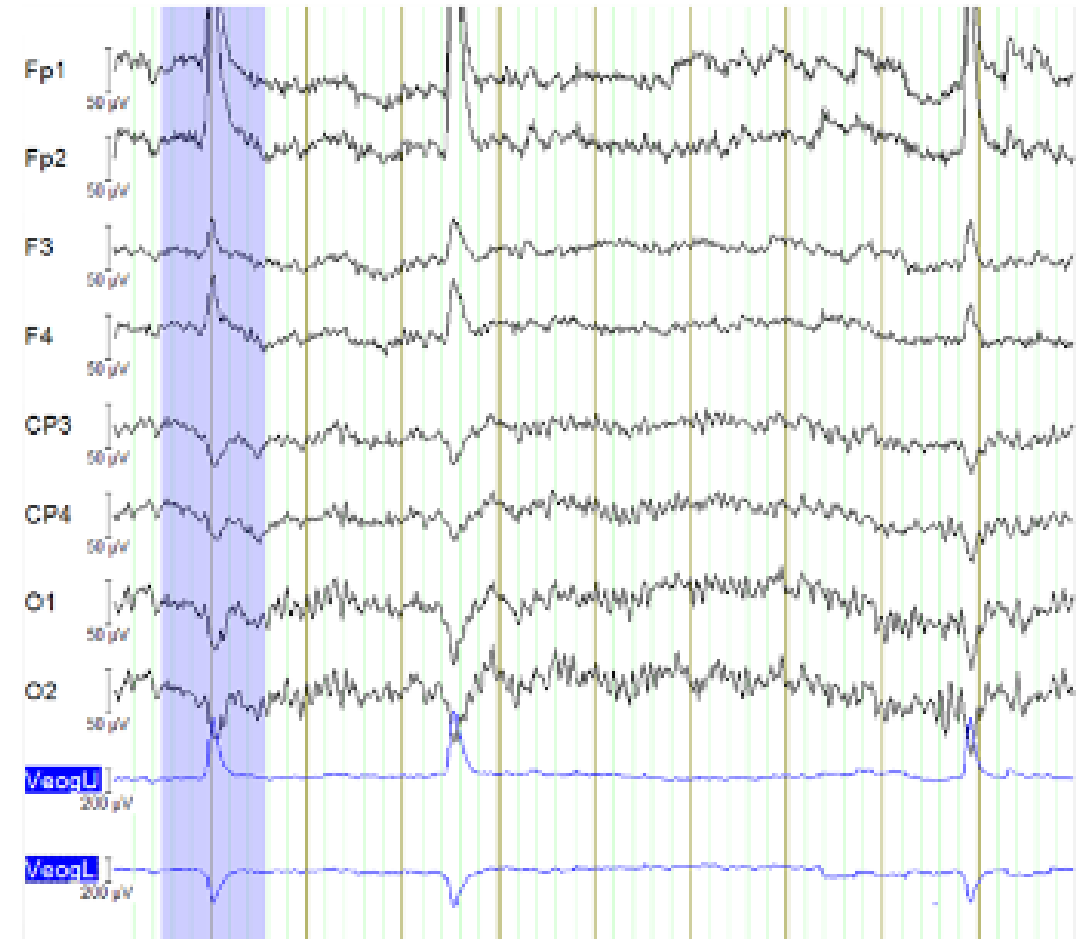
It is especially suited for removing occasional large-amplitude artifacts from EEG recordings.

Instead of traditional data cutting or filtering, it reconstructs the noisy part of the signal based on its other fragments.

Visualizing – Common EEG artefacts – Eye blinks

The artifact is especially noticeable at frontal channels near the eyes, reaching over 100 microvolts.

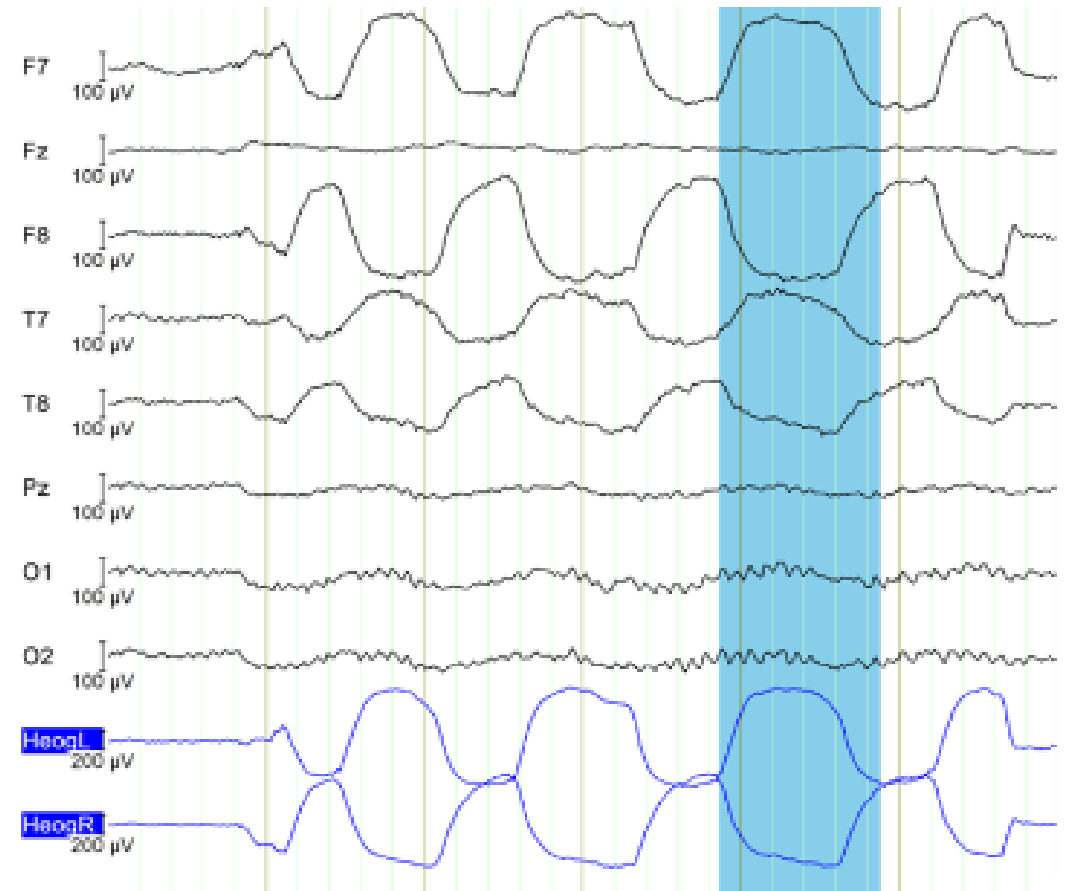
Most of its frequencies are in the EEG delta and theta bands.



Visualizing – Common EEG artefacts – Eye movements

Most prominent over channels near the side of the head behind the eye, but also influence channels elsewhere.

Eye movements form a box shape that peaks in the delta and theta bands but affects up to 20 Hz.

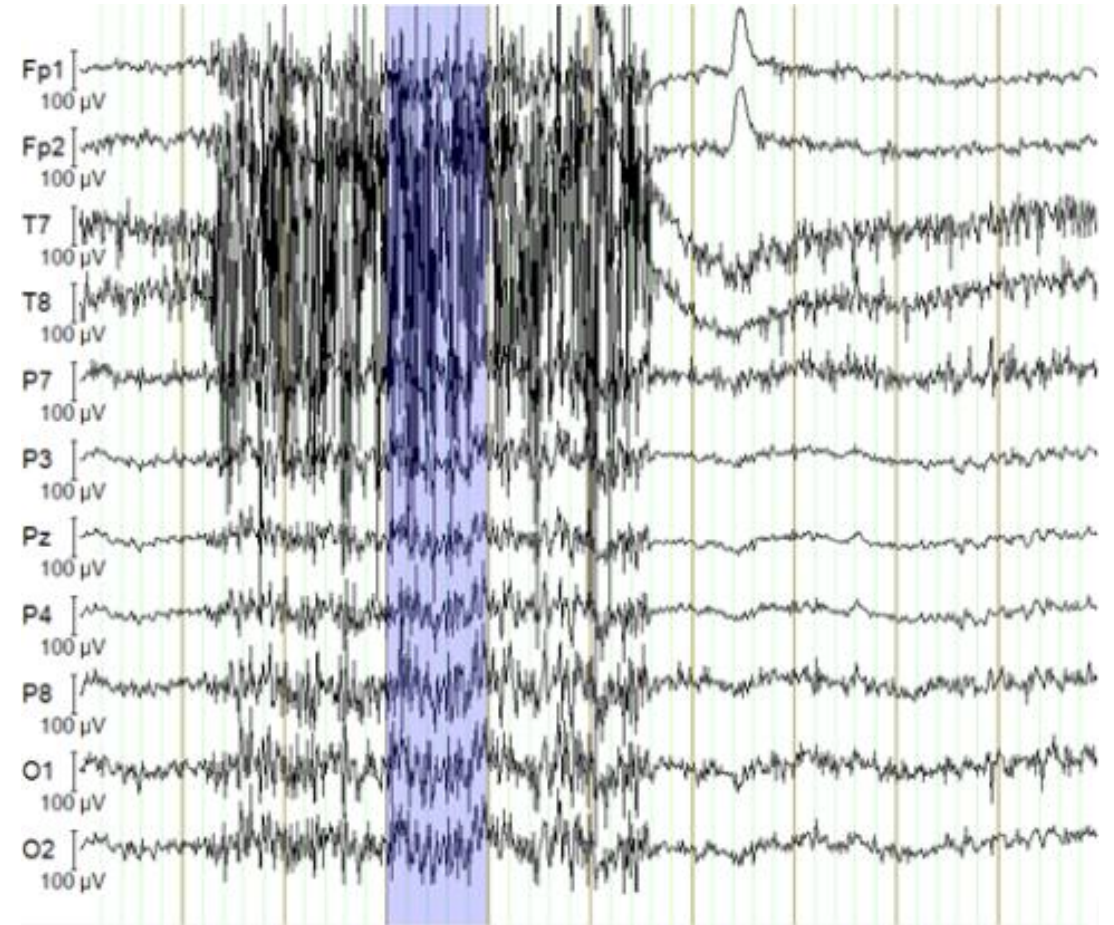


Visualizing – Common EEG artefacts – Muscular artefacts

Activation of muscles creates high frequencies that span the EEG range. The muscle groups and activation intensity determine their amplitude.

Activation with teeth clenching makes a strong scalp-wide noise. Other head muscles like the jaw and forehead make smaller artifacts.

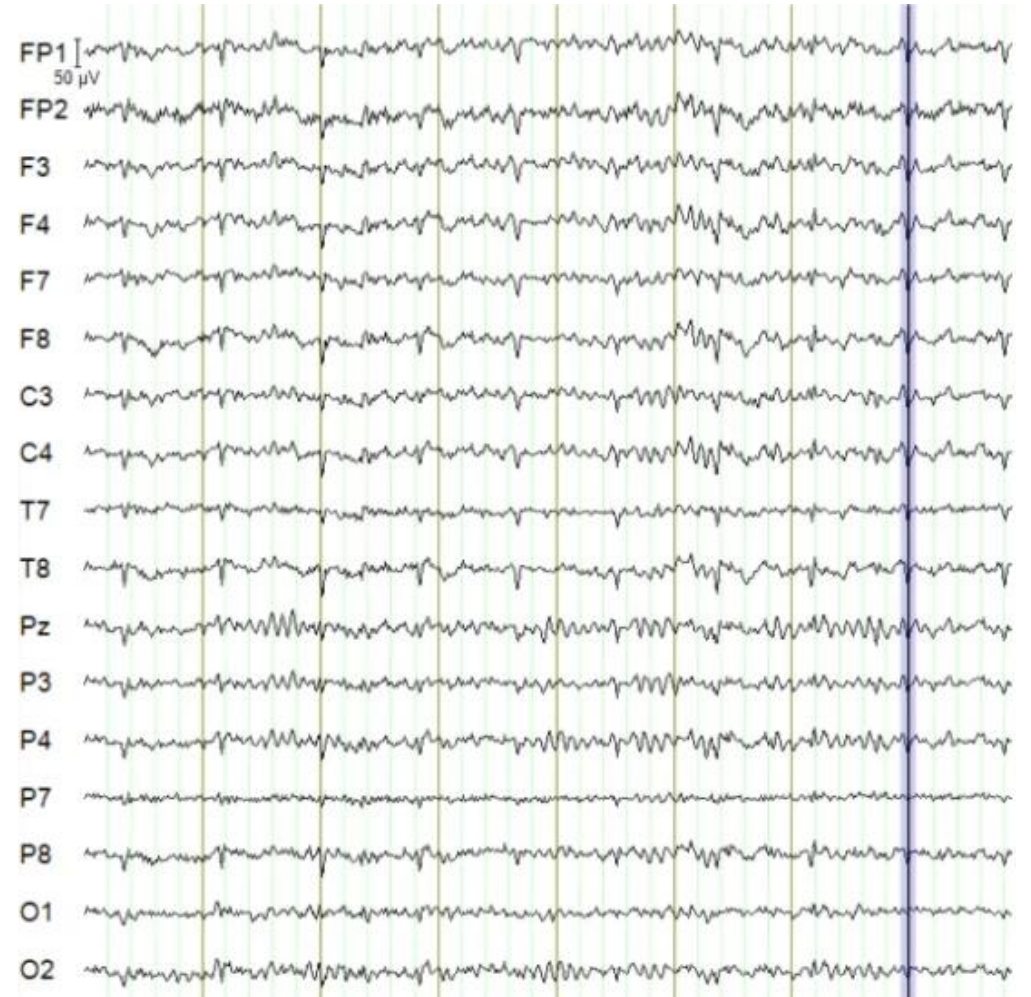
Muscular artifacts influence all EEG frequencies and are particularly noticeable above 20 Hz and up to 300 Hz.



Visualizing – Common EEG artefacts – Pulse artifacts

Head artery pulsation can create modest rhythmic movement of electrodes.

It can be mistaken for EEG rhythms due to their small size, causing time and frequency domain confounds.



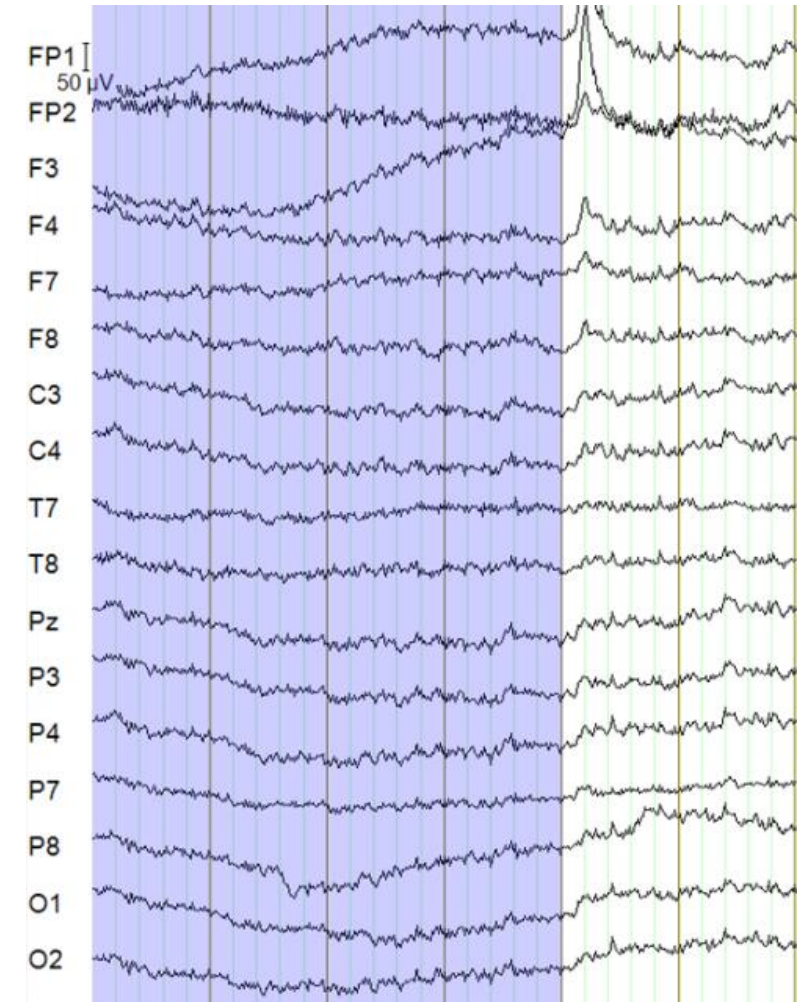
Visualizing – Common EEG artefacts – Sweating

Even slight sweating alters skin conductivity. This artifact is caused by heat, exercise, or stress.

The scalp voltage drifts slowly in the recording. Drifts vary in frequency and amplitude.

These drifts influence time-domain signal timing and amplitude.

Power is mostly at slow frequencies in this artifact (below 1 Hz).



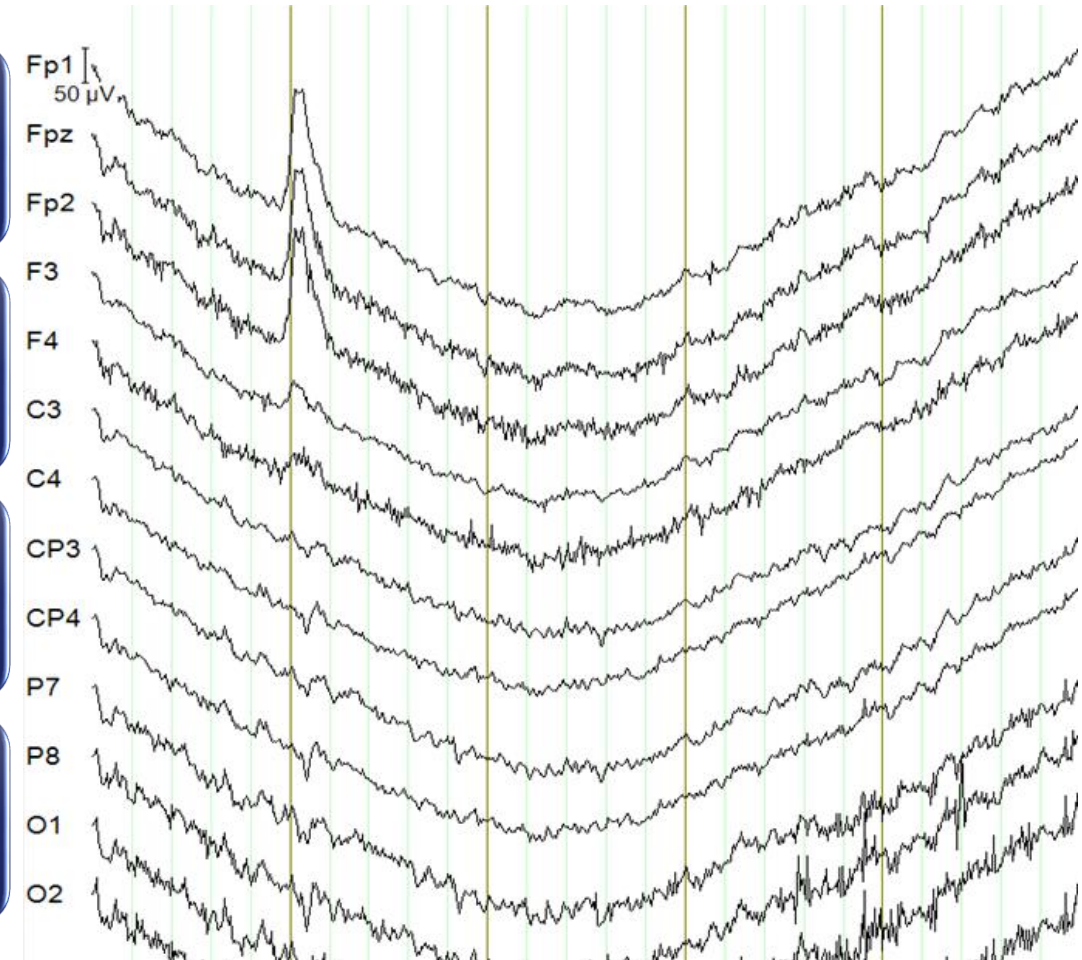
Visualizing – Common EEG artefacts – Body movements

Movement causes the EEG cap to slightly move over the scalp, especially when loose.

Gross movements cause huge voltage shifts that can briefly saturate amplifiers.

While channel drift is gradual with minor body movements.

Complex task movements cause cap pulling, sliding, and shaking that affect all channels.



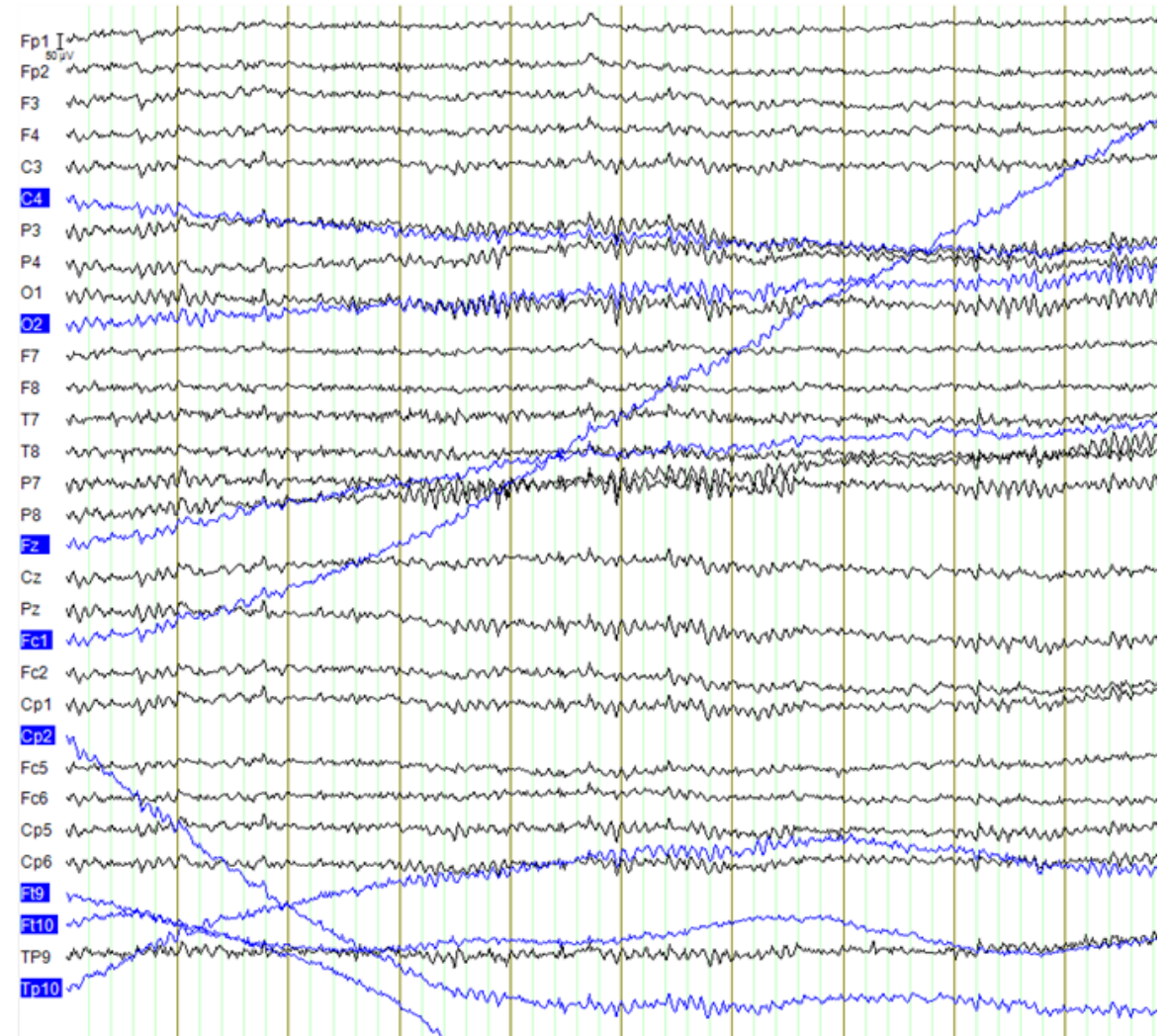
Visualizing – Common EEG artefacts – Loose electrode contact

When the scalp-electrode contact is broken, the signal is unsteady. Hair, body movement, or a loose cap might cause this.

Slow signal drifts are common with loose contact.

However, electrochemical instabilities caused by loose contact might cause rapid conductance shifts and a “electrode pop” in the data.

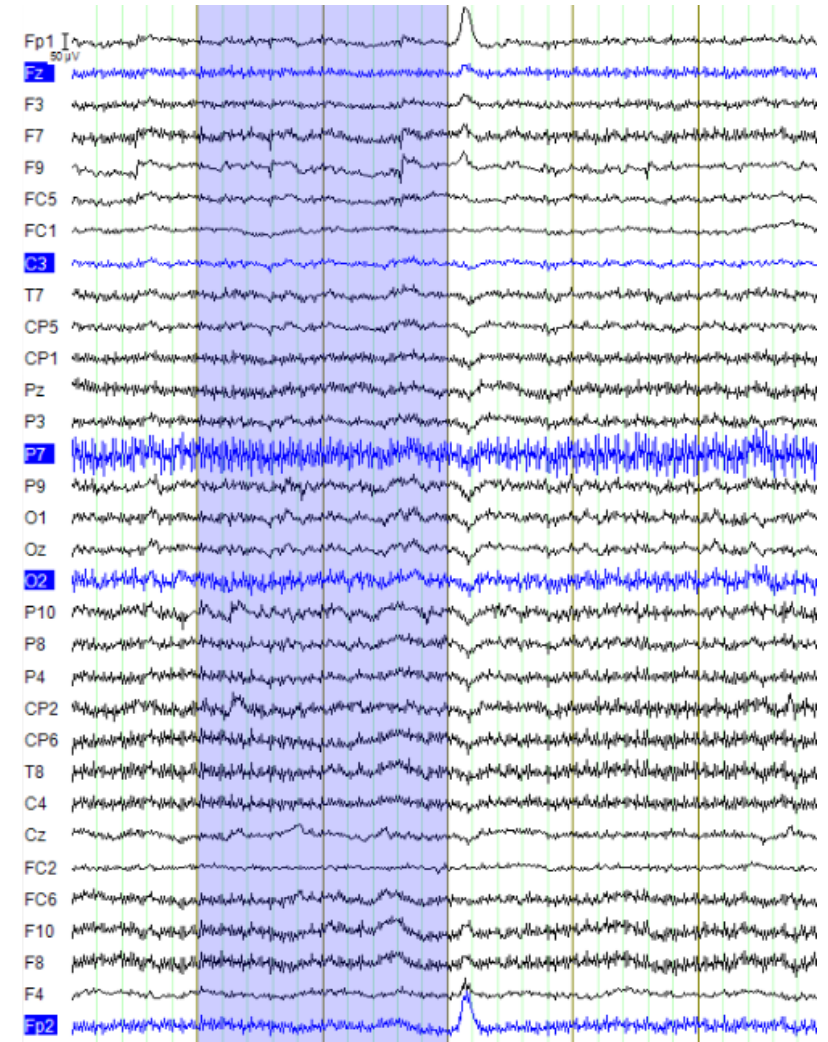
The artifact can also damage ground and reference electrodes, affecting all channels.



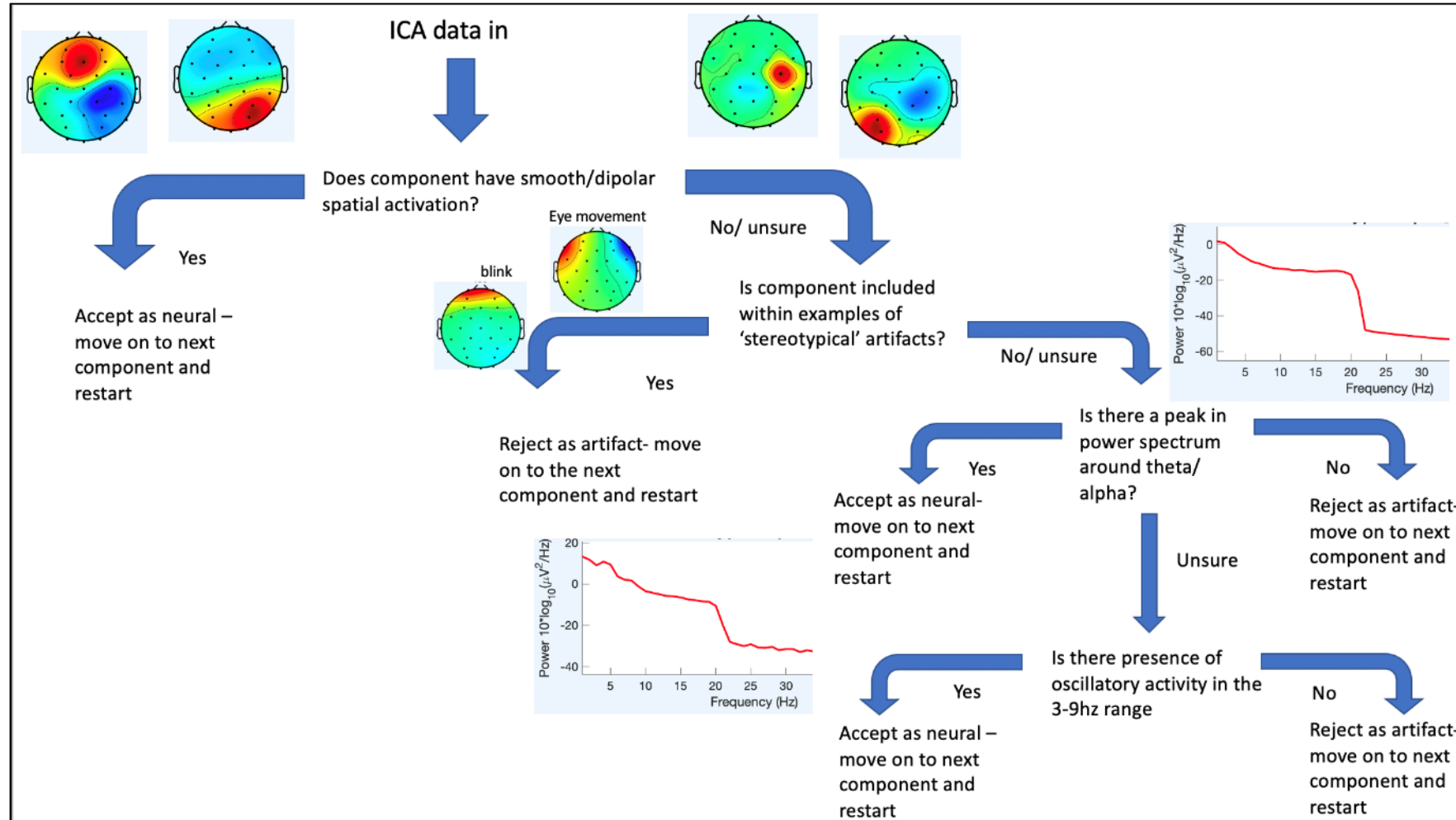
Visualizing – Common EEG artefacts – Line noise

As alternating current travels through the room's electric wiring, it causes electromagnetic noise termed "line noise".

It is picked up by the EEG wires with a frequency that depends on the local grid: 50 or 60 Hz, depending on the country.



Preprocessing of common EEG artefacts – ICA denoising



Quality control of pre-processing – Best practise

Before data collection

System similarity

Procedure
checklist

During data collection

Artifact
avoidance

Unavoidable
limitations

How can you check the quality of your analysis?



Pre-processing open-source software

EEGLAB



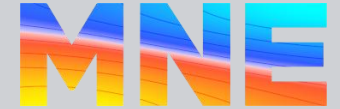
- **Language:**
 - MATLAB
- **Strengths:**
 - Extensive toolbox with a wide range of features for preprocessing, analysis, and visualization.
 - Large community support and active development.
 - User-friendly interface with a graphical user interface (GUI).
- **Weaknesses:**
 - Can be resource-intensive, especially for large datasets.
 - Requires MATLAB license.

Fieldtrip



- **Language:**
 - MATLAB
- **Strengths:**
 - Highly customizable and flexible for various EEG and MEG analysis tasks.
 - Strong focus on time-frequency analysis and source localization.
 - Efficient handling of large datasets.
- **Weaknesses:**
 - Can have a steeper learning curve due to its flexibility.
 - May require more manual configuration compared to EEGLAB.

MNE

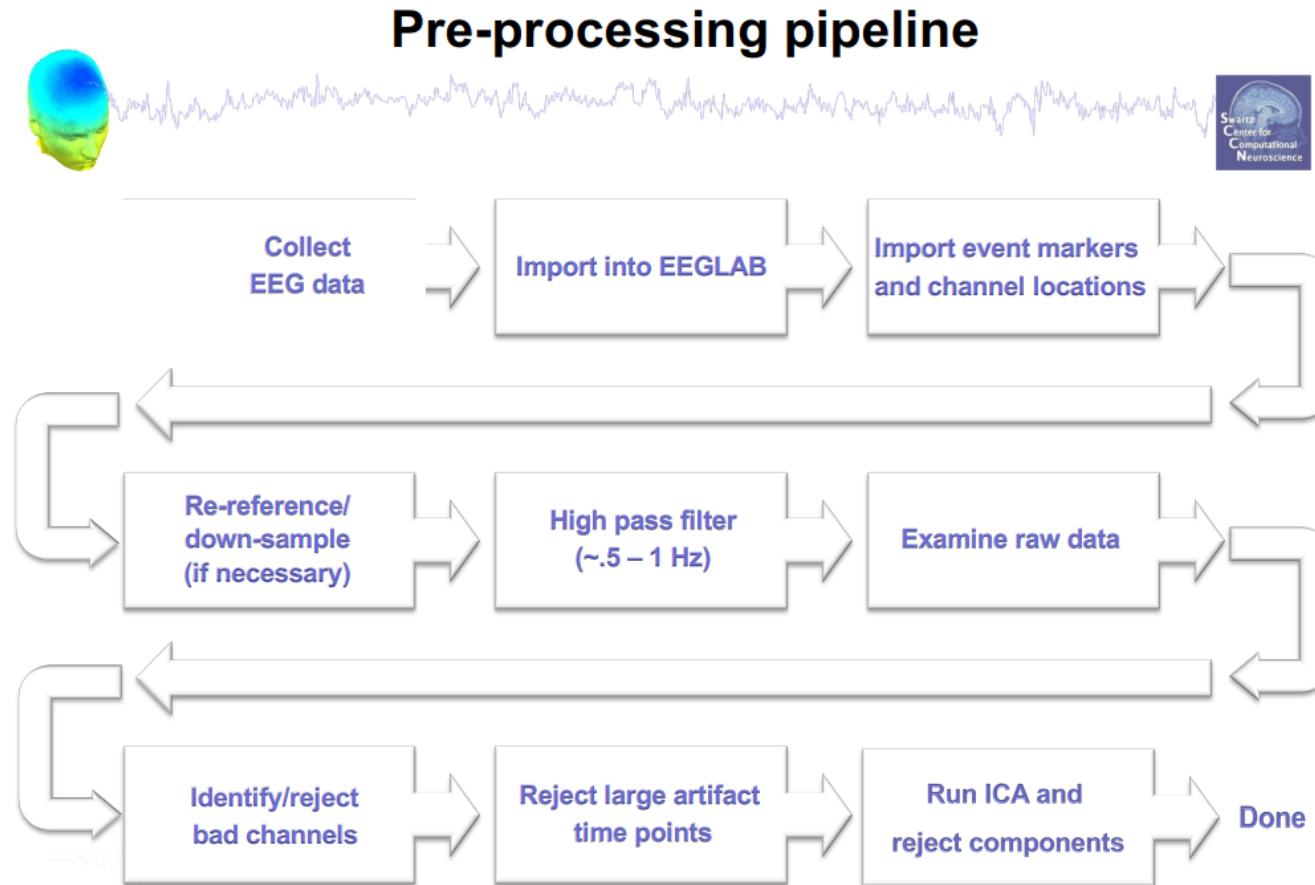


- **Language:**
 - Python
- **Strengths:**
 - Powerful and flexible framework for both EEG and MEG data analysis.
 - Well-integrated with the Python scientific computing ecosystem.
 - Offers advanced features for source localization and time-frequency analysis.
- **Weaknesses:**
 - May require more programming knowledge compared to EEGLAB.
 - While the community is growing, it might not be as extensive as EEGLAB's.

Offline pre-processing of EEG data in MATLAB

Handling and denoising

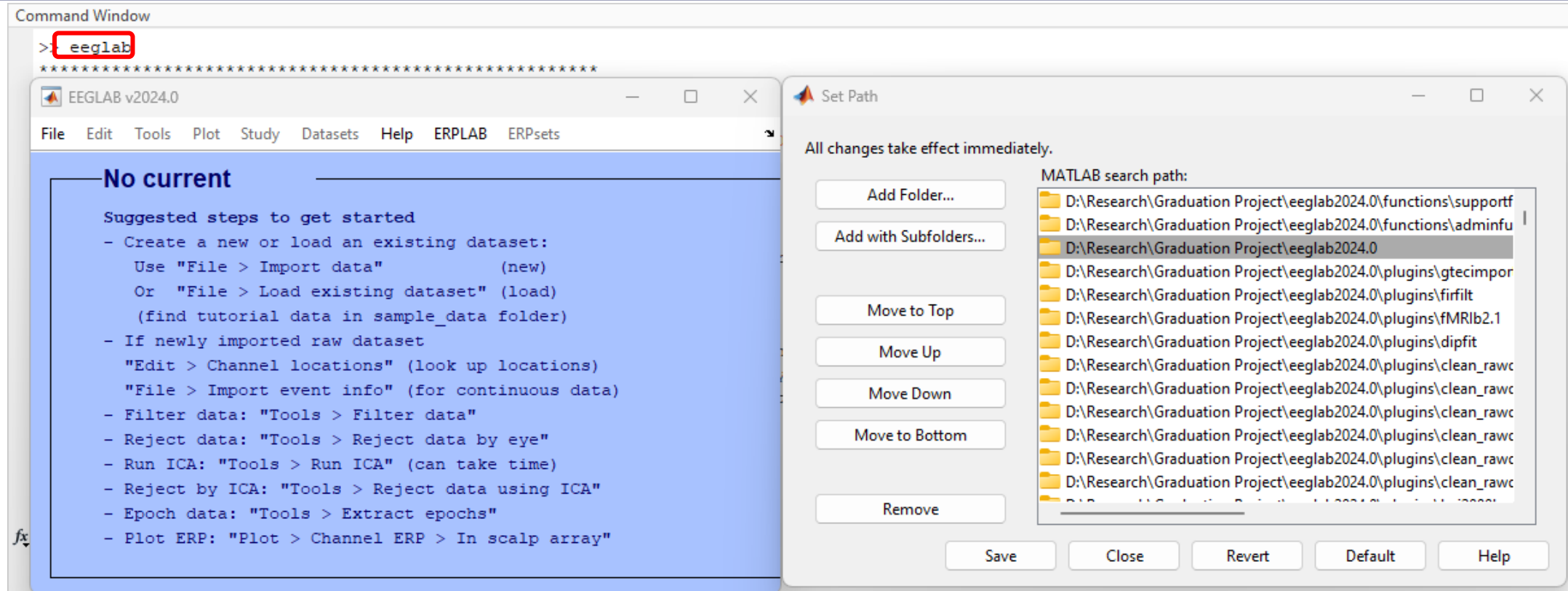
Preprocessing pipeline using EEGLAB



Start EEGLAB through MATLAB

Add EEGLAB to MATLAB path.

Open MATLAB, then type 'eeglab' in the command window.



Import EEG data into EEGLAB

From the EEGLAB GUI:

- **File>Load existing dataset**

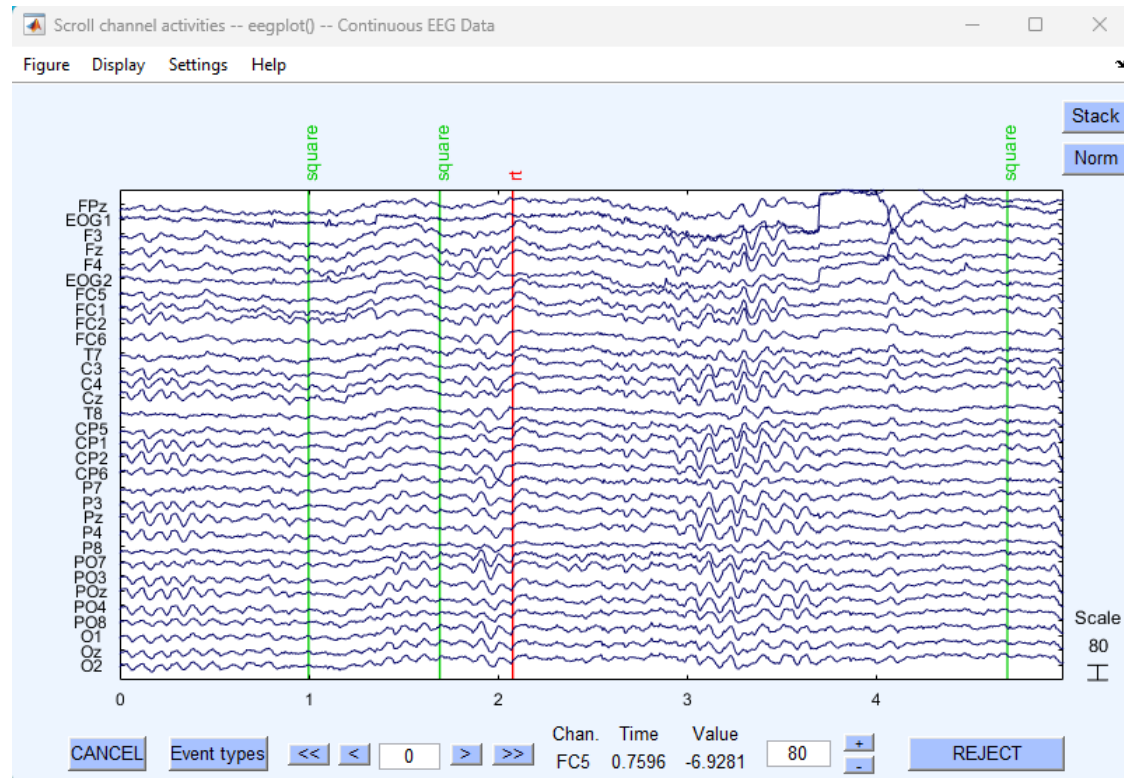
There are multiple options on how to load a file depending on the acquisition system used.

Note: You might need to install some plugins to EEGLAB which are not pre-installed.

Plot raw EEG data

Checking data after each step is a common practice, to view the data:

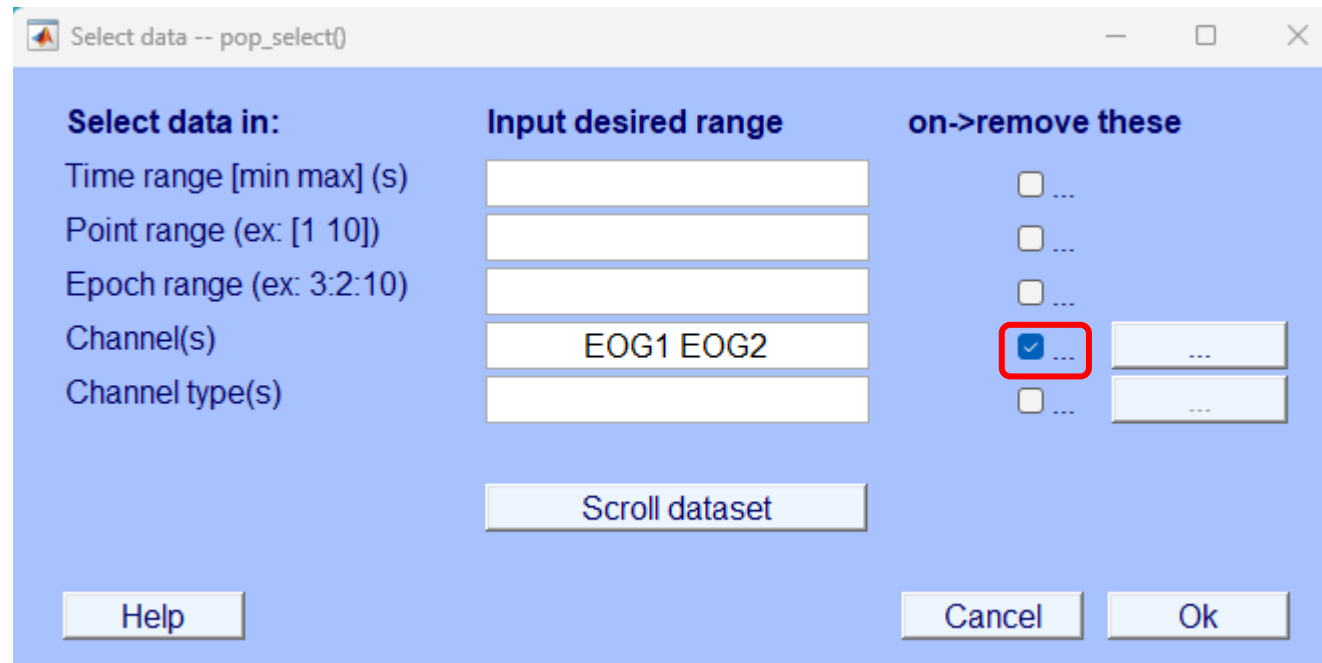
- **Plot>Channel data (scroll)**



Delete channels (if needed)

To do that go for:

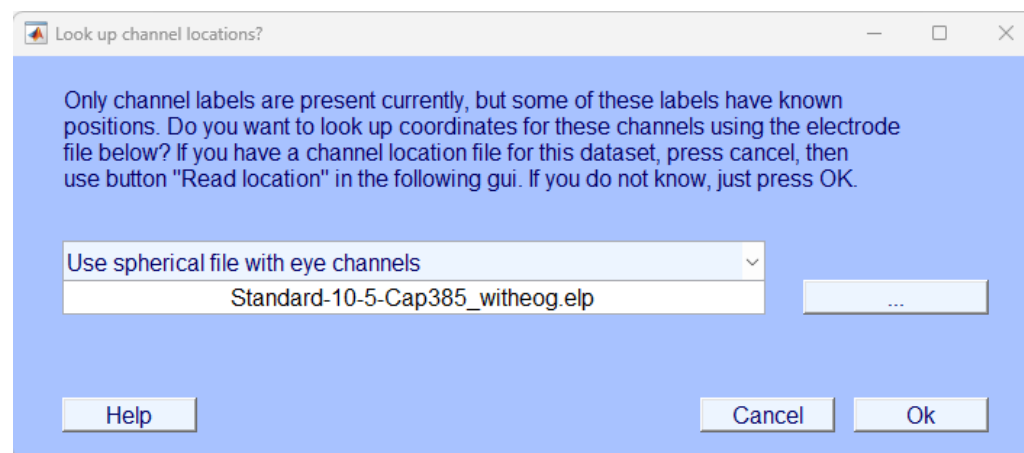
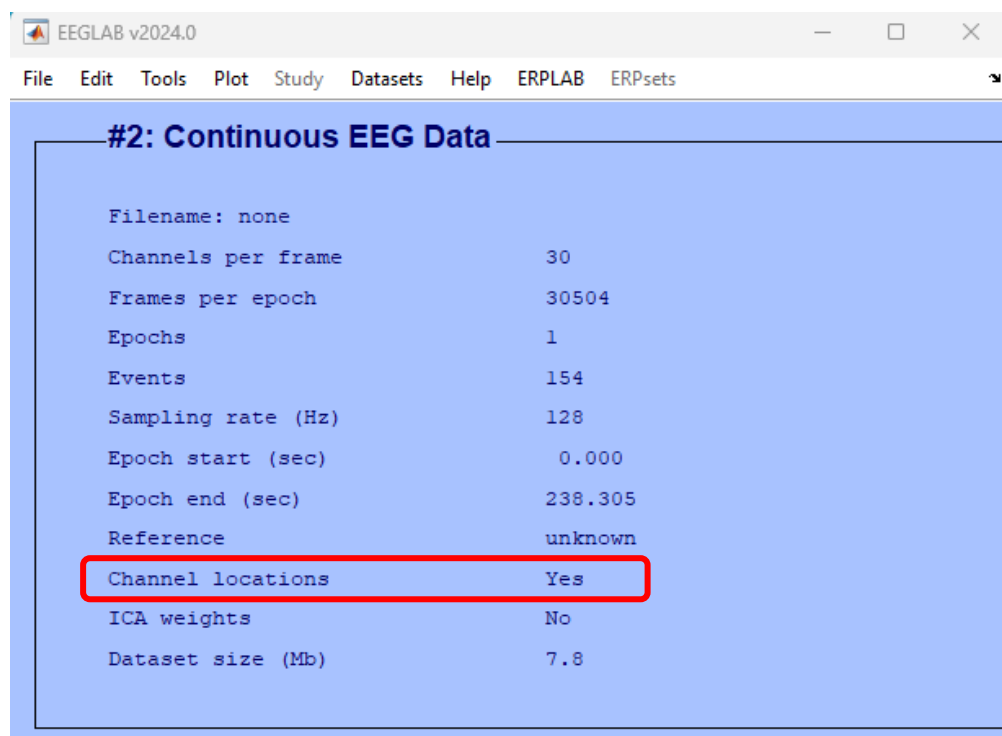
- **Edit>Select data**



Import channel locations (if needed)

To do that go for (if needed):

- **Edit>Channel locations**
- EEGLAB comes with standard montages, pick the one that suits your data.



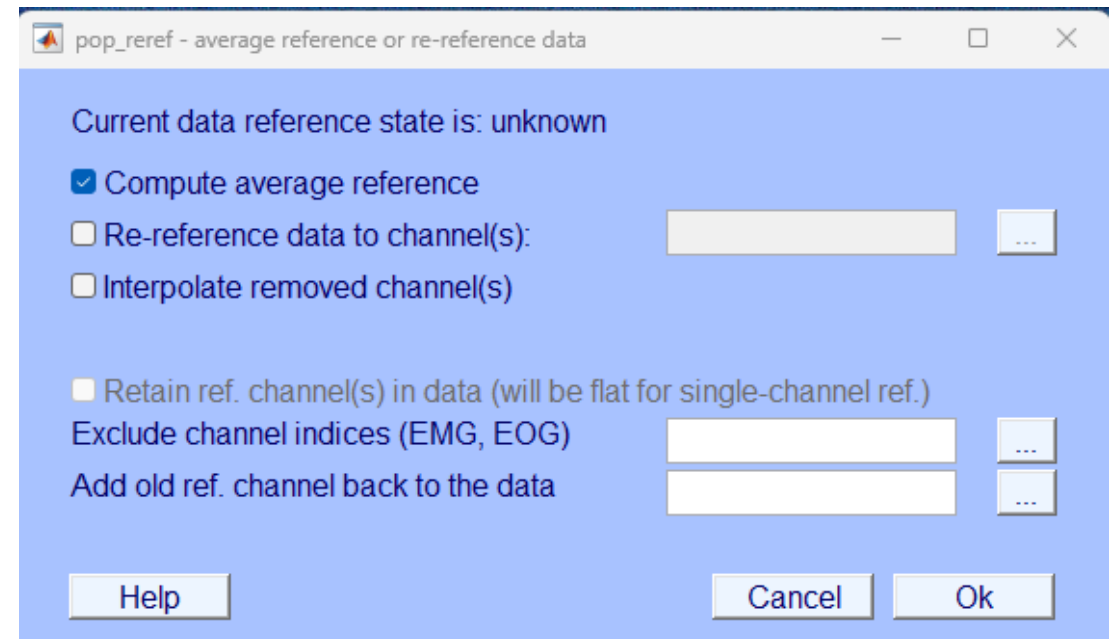
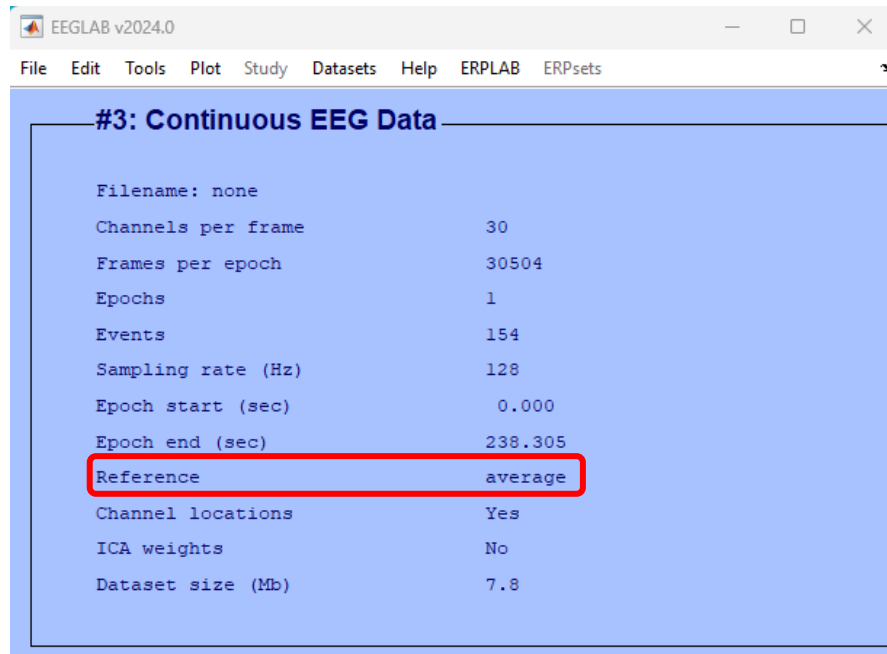
Re-reference

This step is needed to have a sense of electric fields with respect to a reference

- **Tools>Re-reference the data**

We only depend on EEG channels to calculate the common average reference.

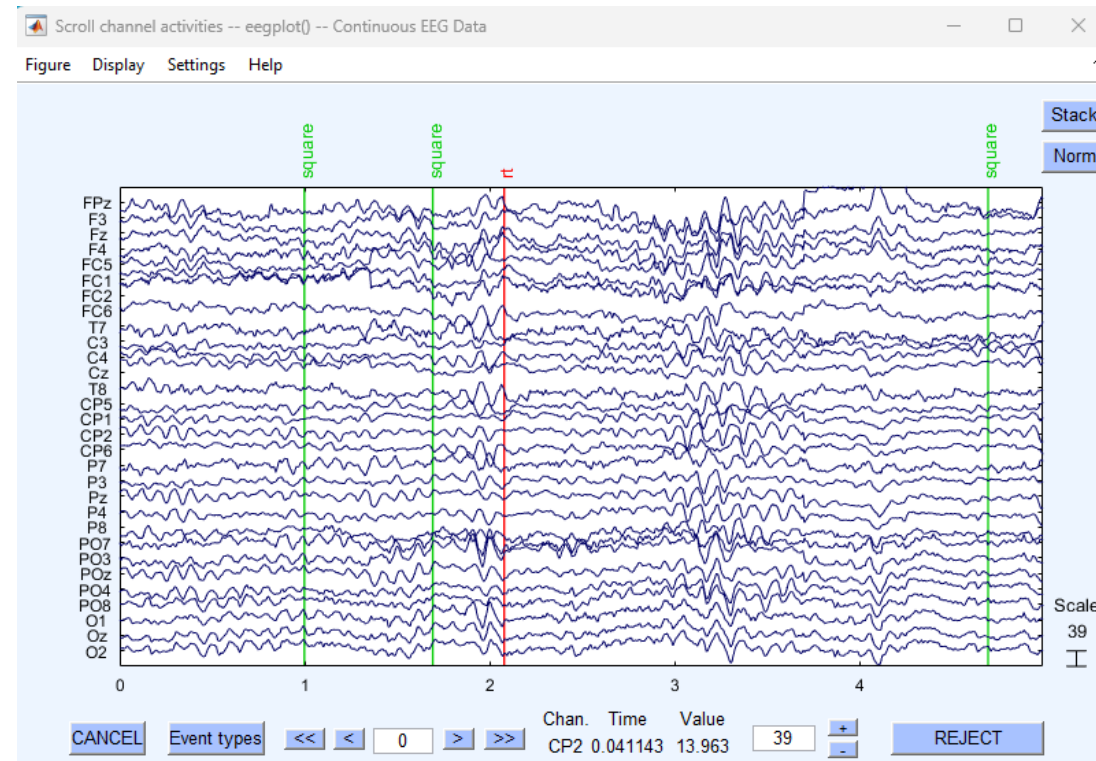
- This one is based on a theory that the brain is a closed electric system.



Plot EEG data after re-referencing

Checking data after each step is a common practice, to view the data:

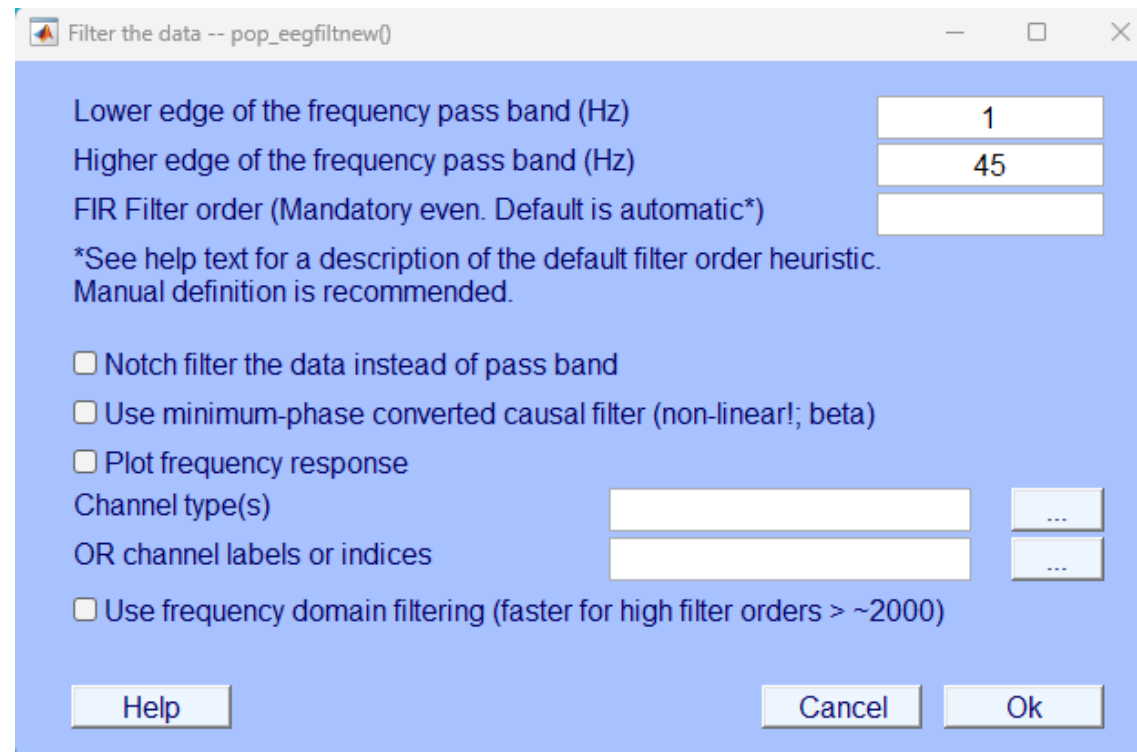
- **Plot>Channel data (scroll)**



Temporal filtering

Filtering is important to specify the range of interest and remove noise.

- **Tools>filter the data>Basic FIR filter**



The screenshot shows a dialog box titled "Filter the data -- pop_eegfiltnew()". It contains several input fields and checkboxes. The "Lower edge of the frequency pass band (Hz)" is set to 1, and the "Higher edge of the frequency pass band (Hz)" is set to 45. The "FIR Filter order (Mandatory even. Default is automatic*)" is set to an empty field. Below these fields, there is a note: "*See help text for a description of the default filter order heuristic. Manual definition is recommended." There are three checkboxes: "Notch filter the data instead of pass band" (unchecked), "Use minimum-phase converted causal filter (non-linear!; beta)" (unchecked), and "Plot frequency response" (unchecked). Below the checkboxes, there are two input fields: "Channel type(s)" and "OR channel labels or indices", each followed by a button with three dots "...". At the bottom, there is a checkbox "Use frequency domain filtering (faster for high filter orders > ~2000)" (unchecked). At the very bottom, there are three buttons: "Help", "Cancel", and "Ok".

Filter the data -- pop_eegfiltnew()

Lower edge of the frequency pass band (Hz) 1

Higher edge of the frequency pass band (Hz) 45

FIR Filter order (Mandatory even. Default is automatic*)

*See help text for a description of the default filter order heuristic.
Manual definition is recommended.

☐ Notch filter the data instead of pass band

☐ Use minimum-phase converted causal filter (non-linear!; beta)

☐ Plot frequency response

Channel type(s) ...

OR channel labels or indices ...

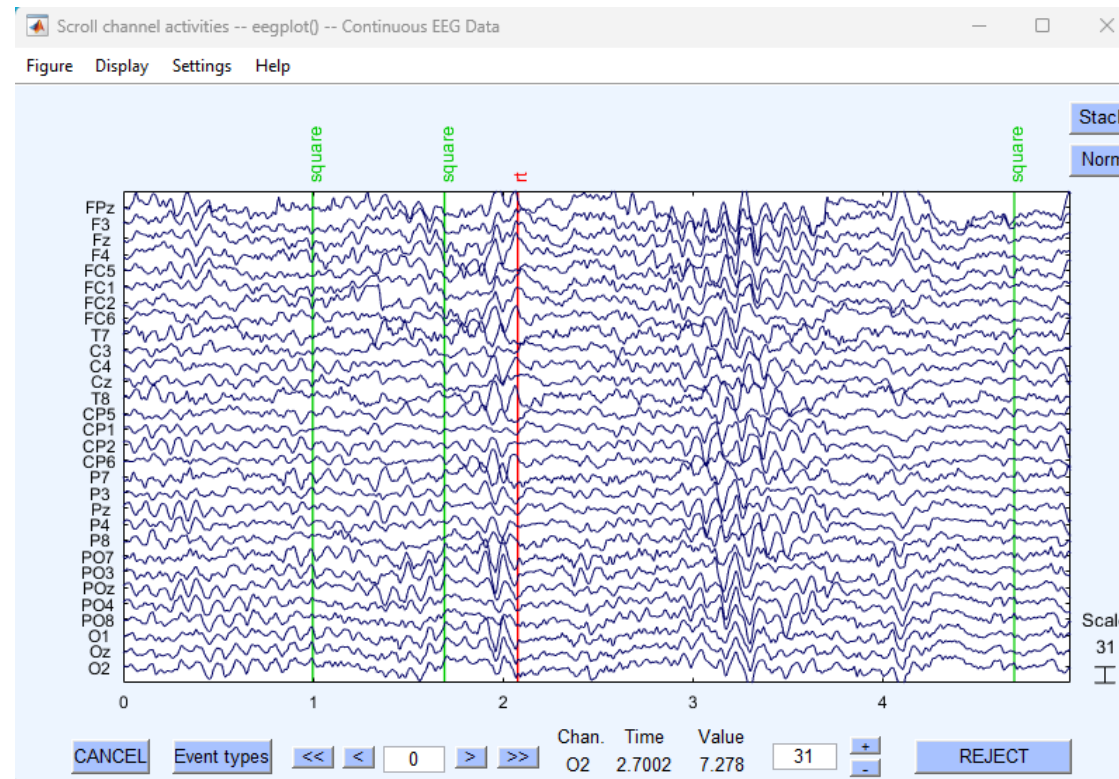
☐ Use frequency domain filtering (faster for high filter orders > ~2000)

Help Cancel Ok

Plot EEG data after filtering

Checking data after each step is a common practice, to view the data:

- **Plot>Channel data (scroll)**



Inspect bad channels

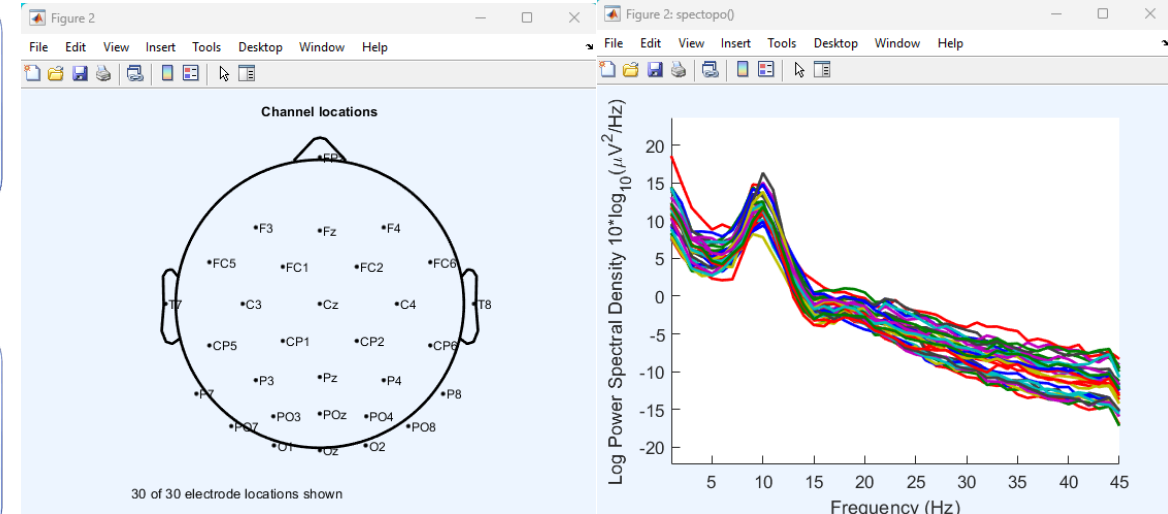
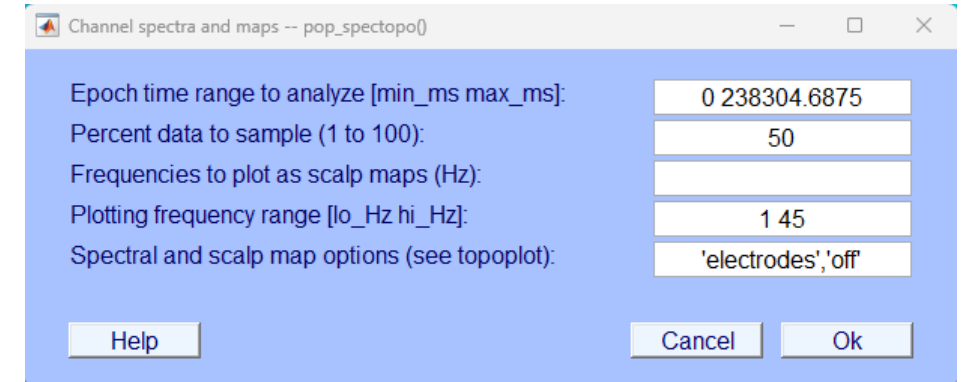
When visually identifying bad channels, you should scroll through the whole dataset, as some channels may only be bad for short periods of time.

- In this case, it might be preferable to remove the corrupted data segment than the channel itself.
- Removing a channel that is only transiently bad is also a trade-off with the number of channels available. When many channels are available, one may easily remove a few channels.

This inspection is done through checking the time series and frequency response of each channel.

- **Plot>Channel spectra and maps**
- **Plot>Channel locations>By name**

Potentially bad channels are either flat or noisy channels in time domain. Another way to identify bad channels is to plot the channels' spectra.



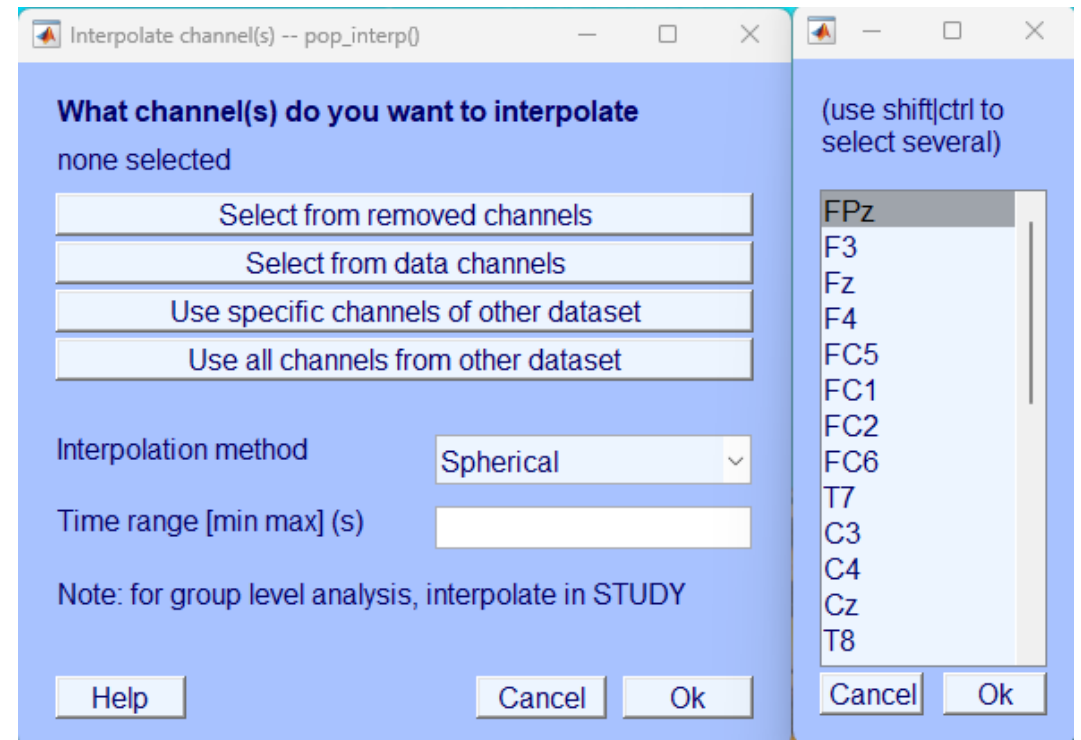
Interpolate bad channels (if needed)

This step is important as filtering alone cannot get rid of noisy channels.

- **Tools>Interpolate electrodes>Select from data channels.**

So, removing bad channels and replacing them by values from their neighbours can solve the problem to keep the number of channels fixed over all subjects.

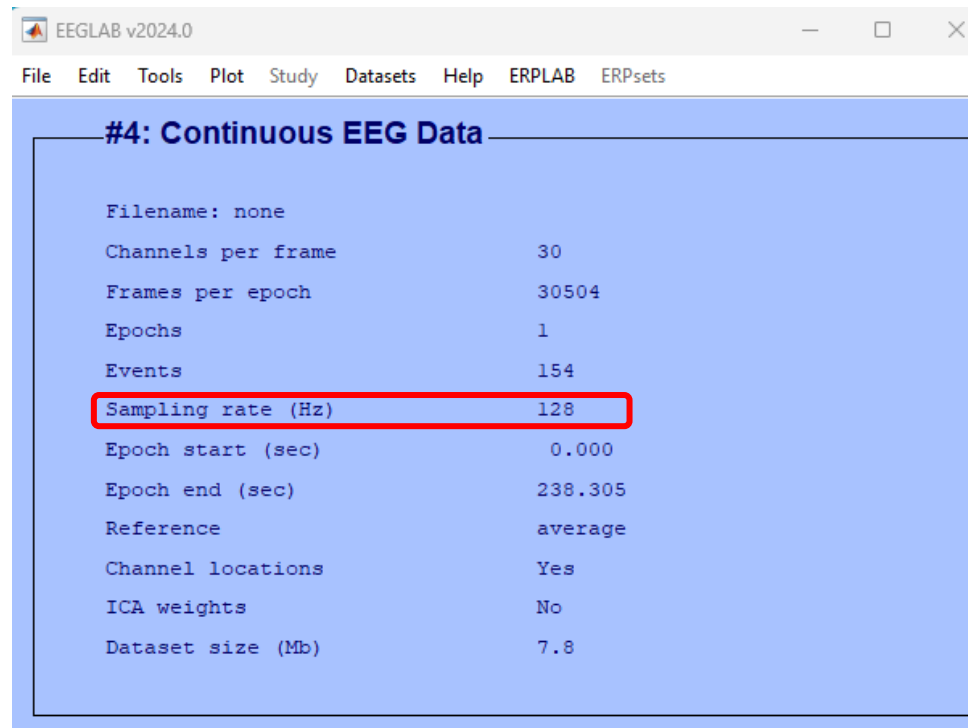
- However, these channels will not carry any new independent data.



Down-sample (if needed)

This step is not mandatory and totally depend on the computational capabilities and what you are looking for in terms of transient change.

- **Tools>Change sampling rate**



Reject large artifacts

This step should be done **before ICA** to get rid of any large artefacts.

- **Tools>Reject data using Clean Rawdata and ASR**

There are several sections to this menu indicating different sequential processes:

- **The top section** is about high-pass filtering your data. By default, this option is not selected since EEGLAB assumes that you might have already filtered your data.
- **The second section** deals with removing bad channels, which was done in previous step.
- **The third section** deals with rejecting bad portions of data using the ASR algorithm.
- **The fourth section** deals with the additional rejection of bad portions of data based on a set number of channels passing a standard deviation threshold in a given time window.
- **The last section** allows plotting results of the rejection with rejected data highlighted.

pop_clean_rawdata()

☐ Remove channel drift (data not already high-pass filtered)
Linear filter (FIR) transition band [lo hi] in Hz: 0.25 0.75

☒ Process/remove channels

- ☐ Only consider these channels
- ☐ Ignore these channels (ECG, EMG, ...)
- ☒ Remove channel if it is flat for more than (seconds): 5
- ☒ Max acceptable high-frequency noise std dev: 4
- ☒ Min acceptable correlation with nearby chans [0-1]: 0.8

☒ Perform Artifact Subspace Reconstruction bad burst correction/rejection

- Max acceptable 0.5 second window std dev: 20
- ☐ Use Riemanian distance metric (not Euclidean) - beta
- ☒ Remove bad data periods (when uncheck, correct using ASR)

☒ Additional removal of bad data periods

- Acceptable [min max] channel RMS range (+/- std dev): -Inf 7
- Maximum out-of-bound channels (%): 25

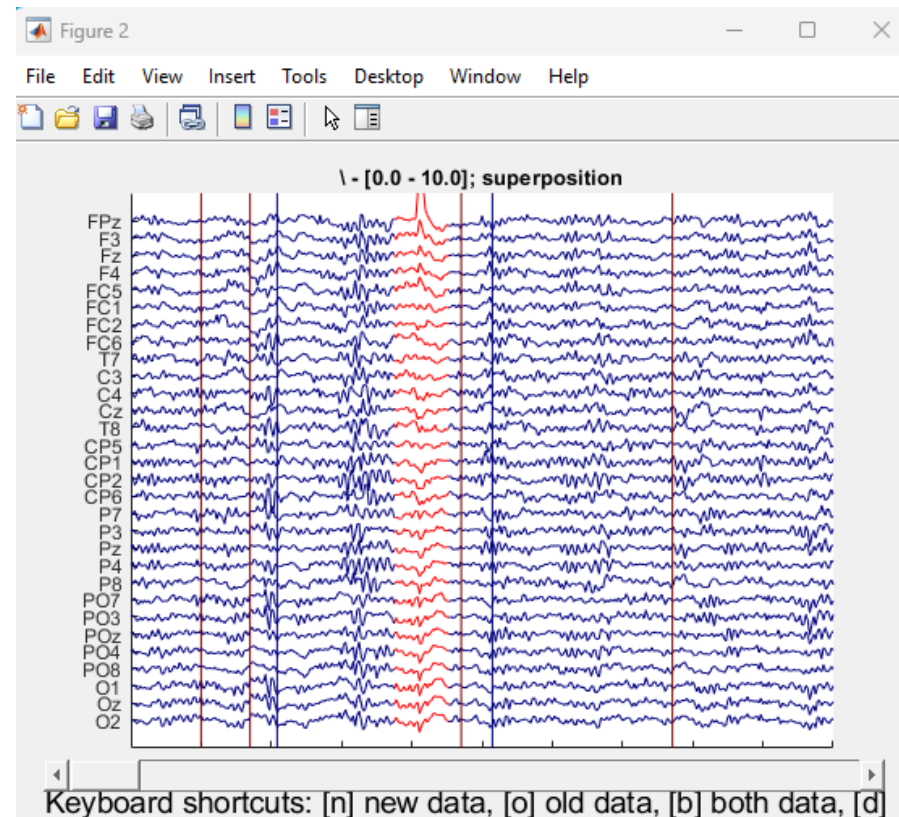
☒ Pop up scrolling data window with rejected data highlighted

Help Cancel Ok

Plot after large artifacts rejection

We can see the rejected channels in red and the rejected data portion when all channels are marked in red in each time segment.

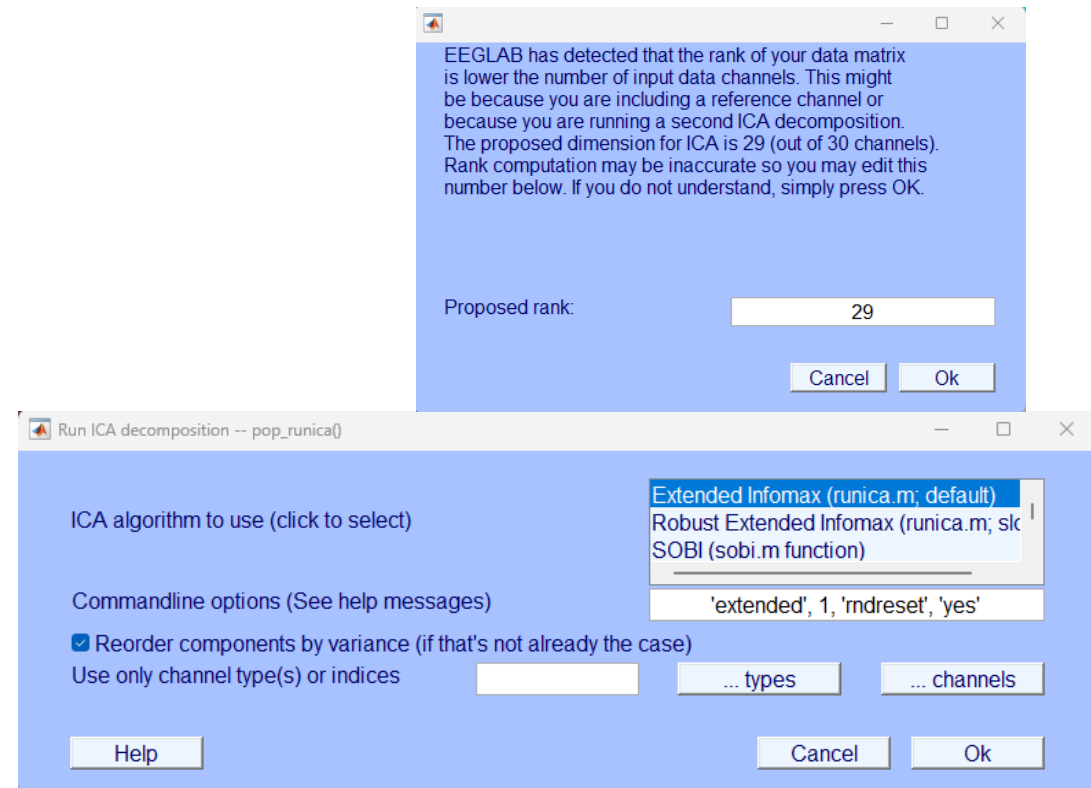
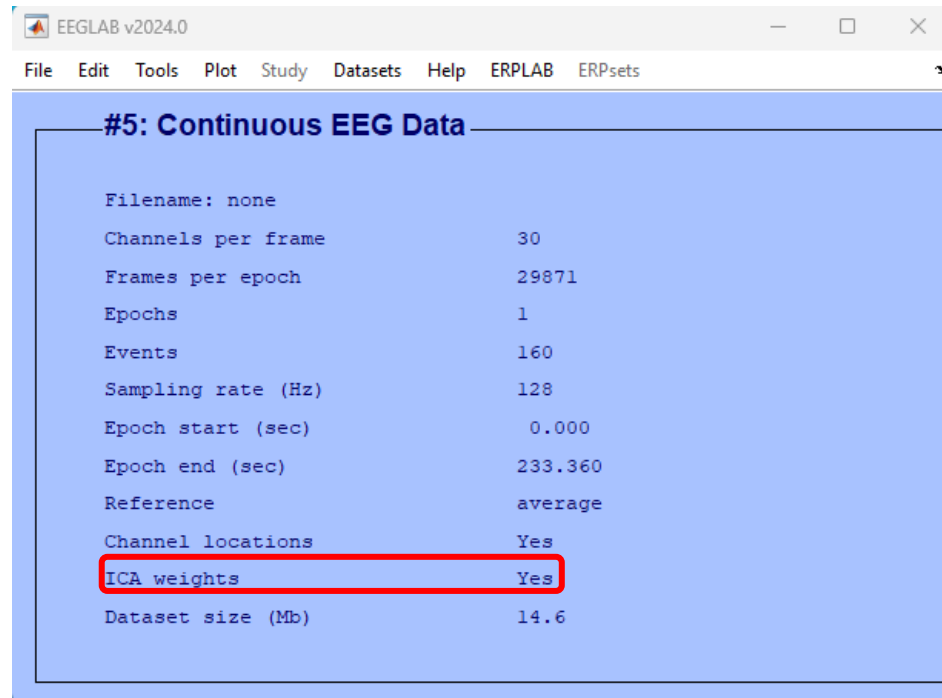
- **Plot>Channel data (scroll)**



Run ICA

This step is important to reject common bad components such eye movements and blinks.

- **Tools>Decompose data by ICA**



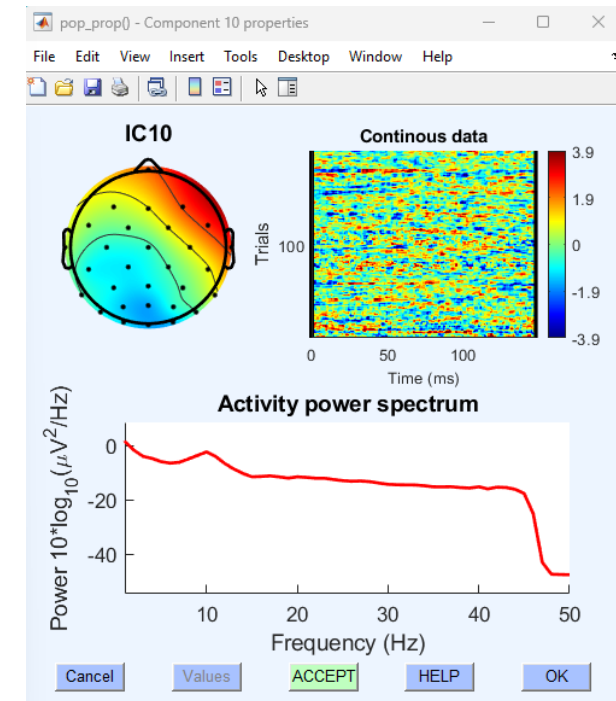
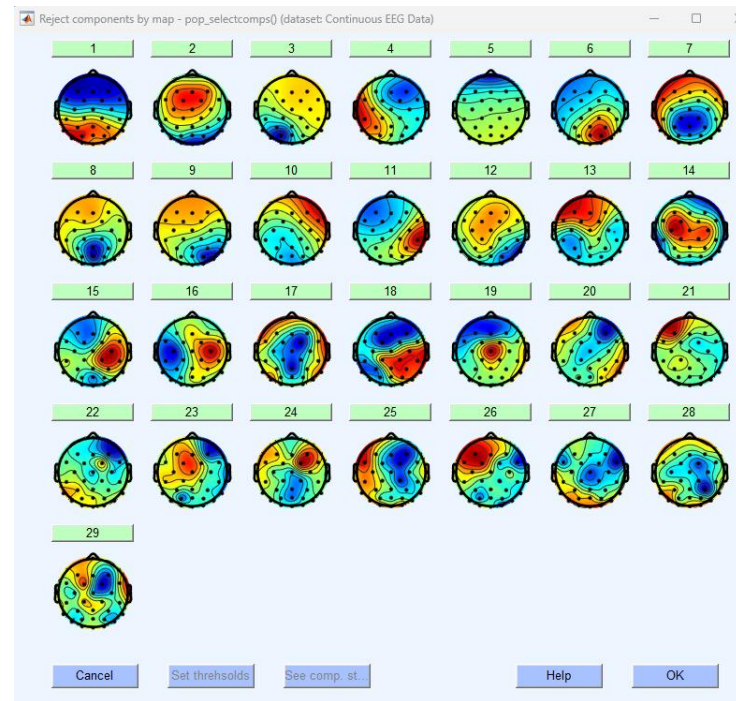
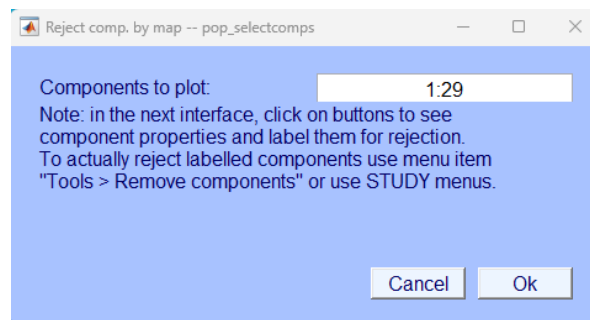
Label bad ICs manually

To label bad ICs manually.

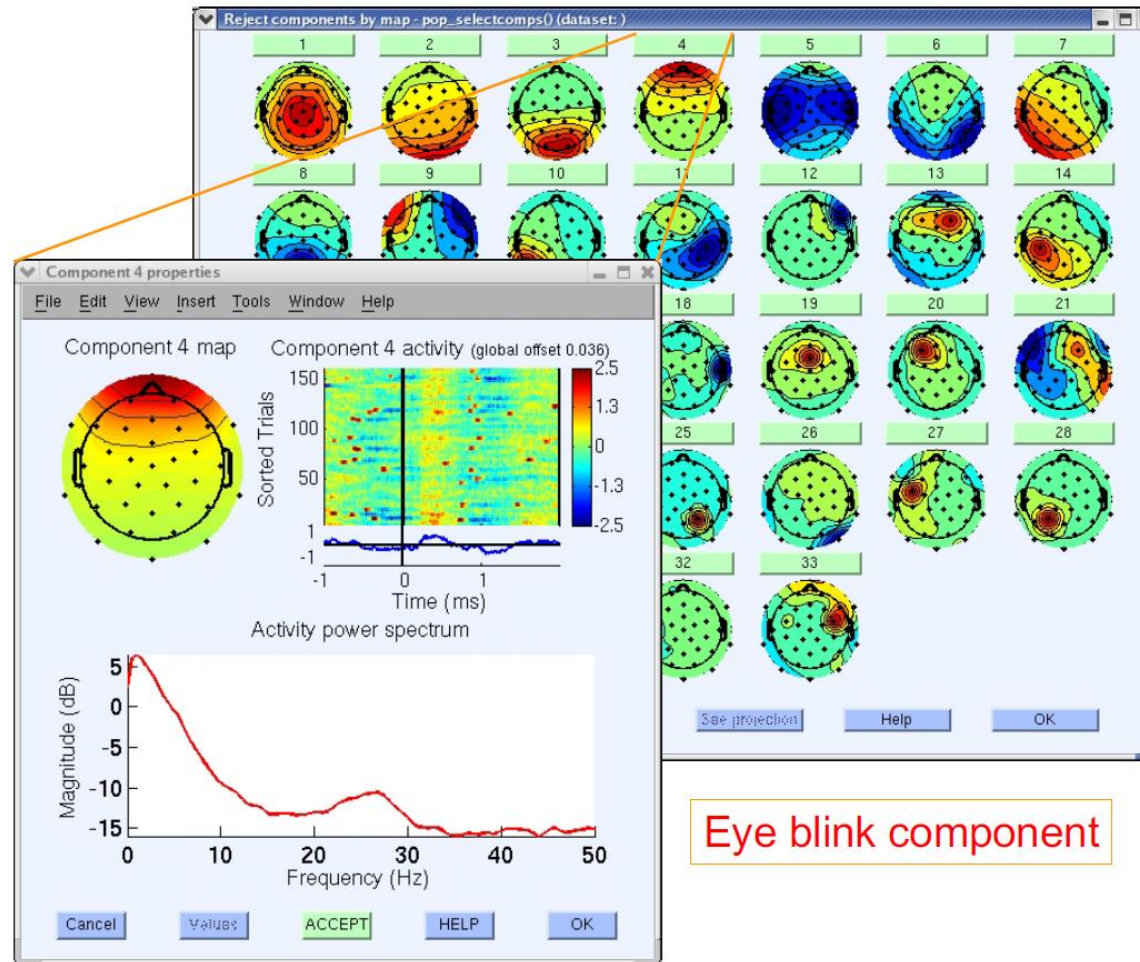
- **Tools>Inspect/label components by map.**

Components with higher variance such as noisy ones will be in the top ones, so plotting the first 15 out of 29 will be enough.

- **You will need to inspect the time course of each IC to confirm.**

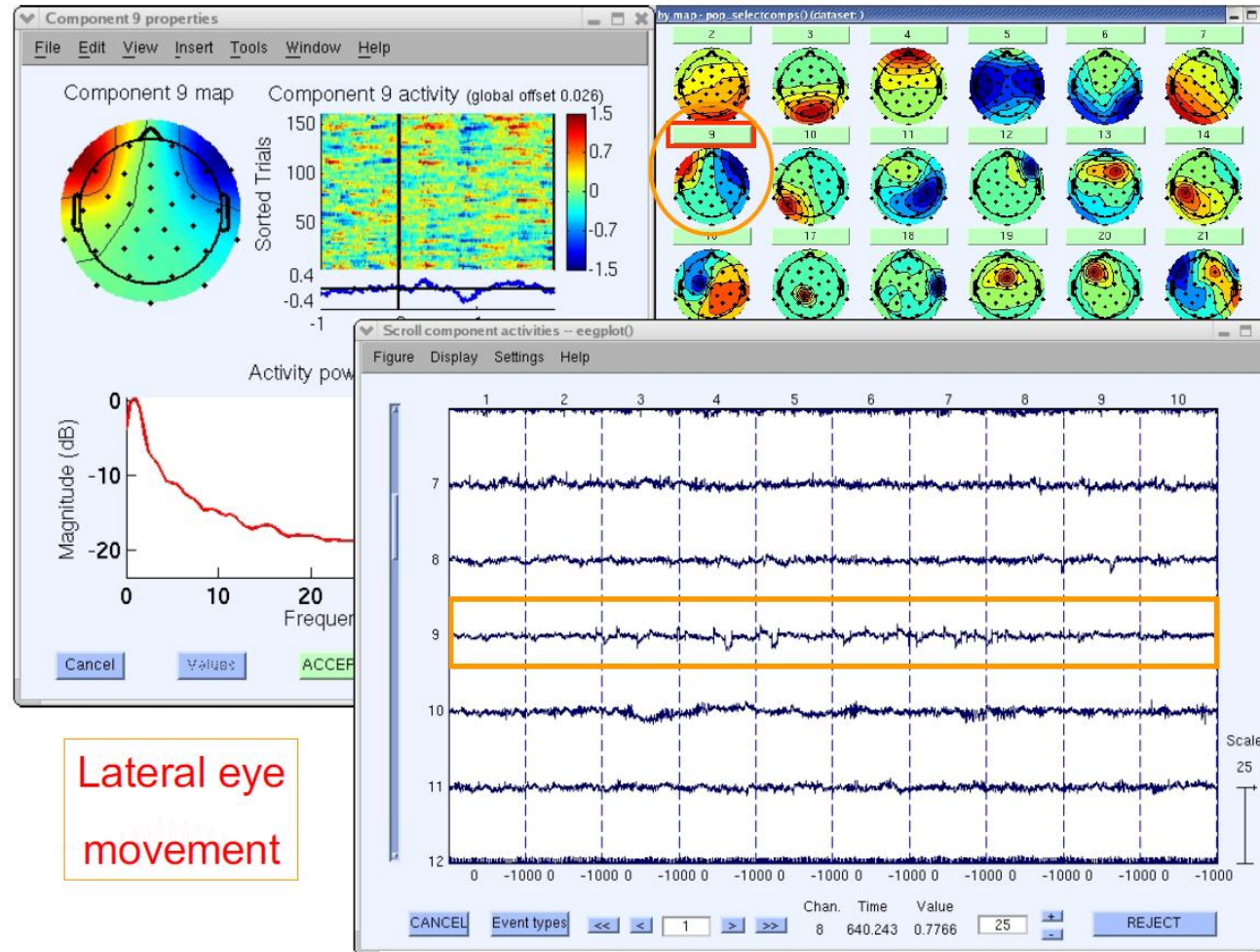


Examples of ICs – Eye blink

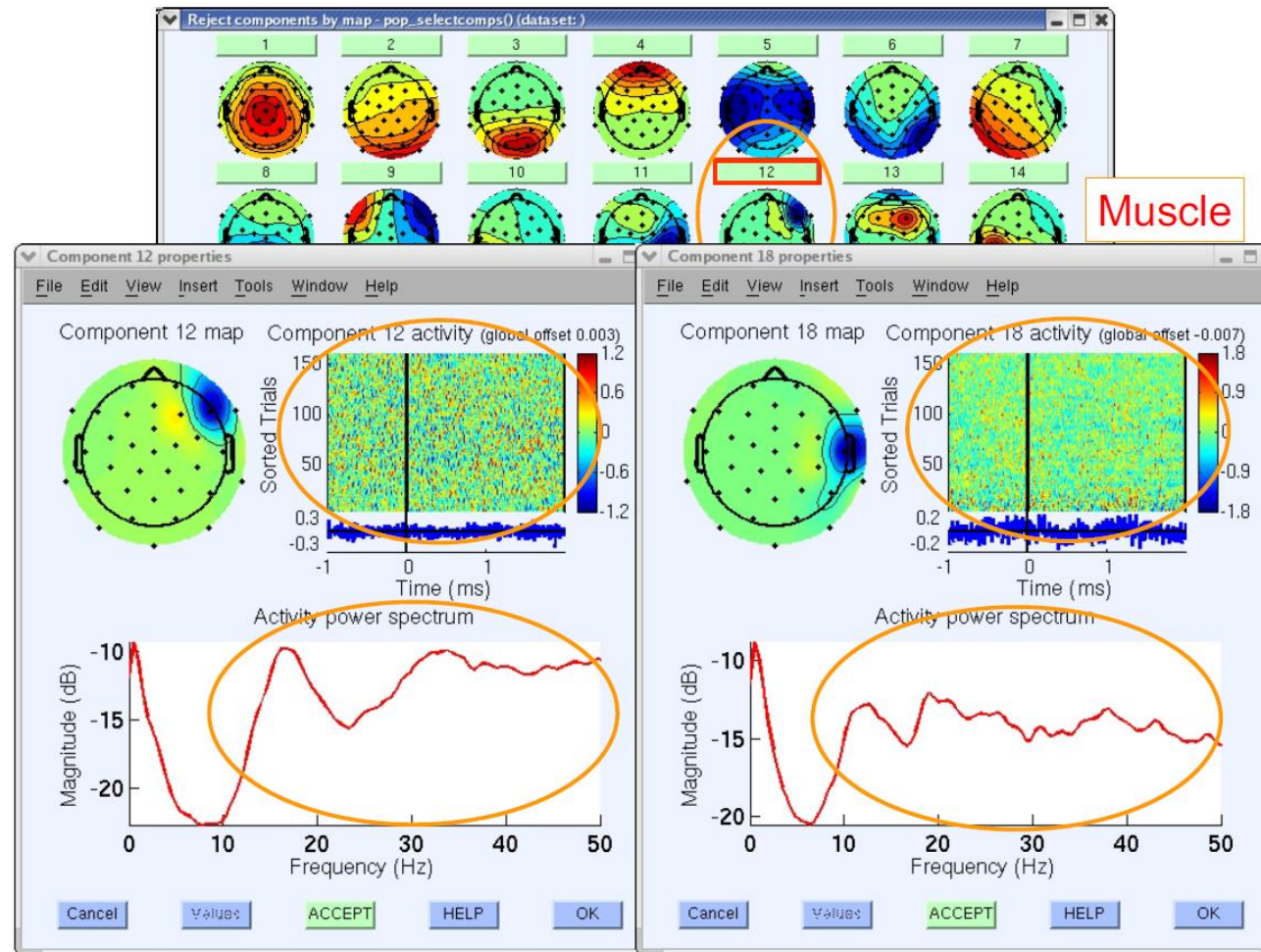


Eye blink component

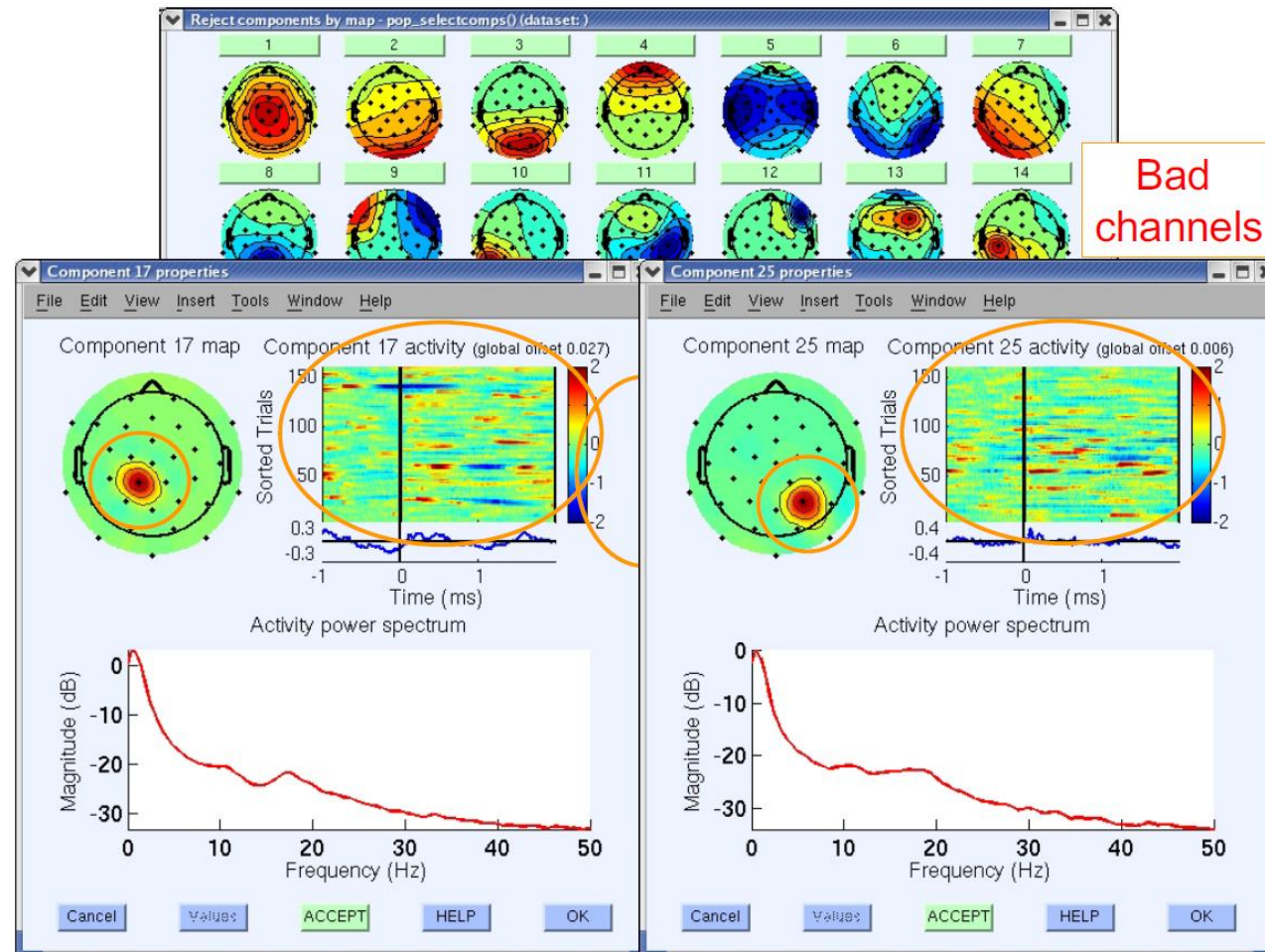
Examples of ICs – Eye movement



Examples of ICs – Muscle

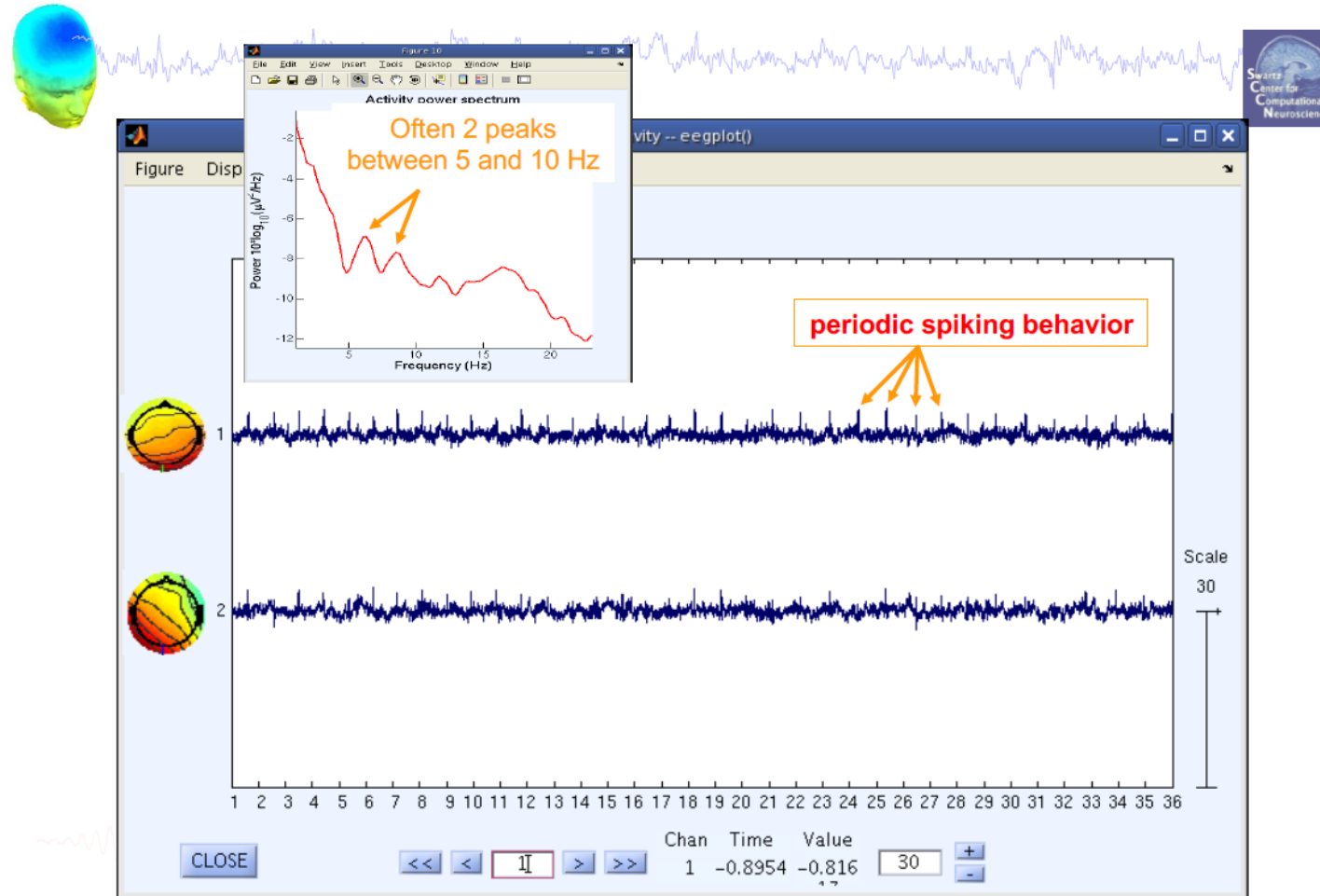


Examples of ICs – Bad channel

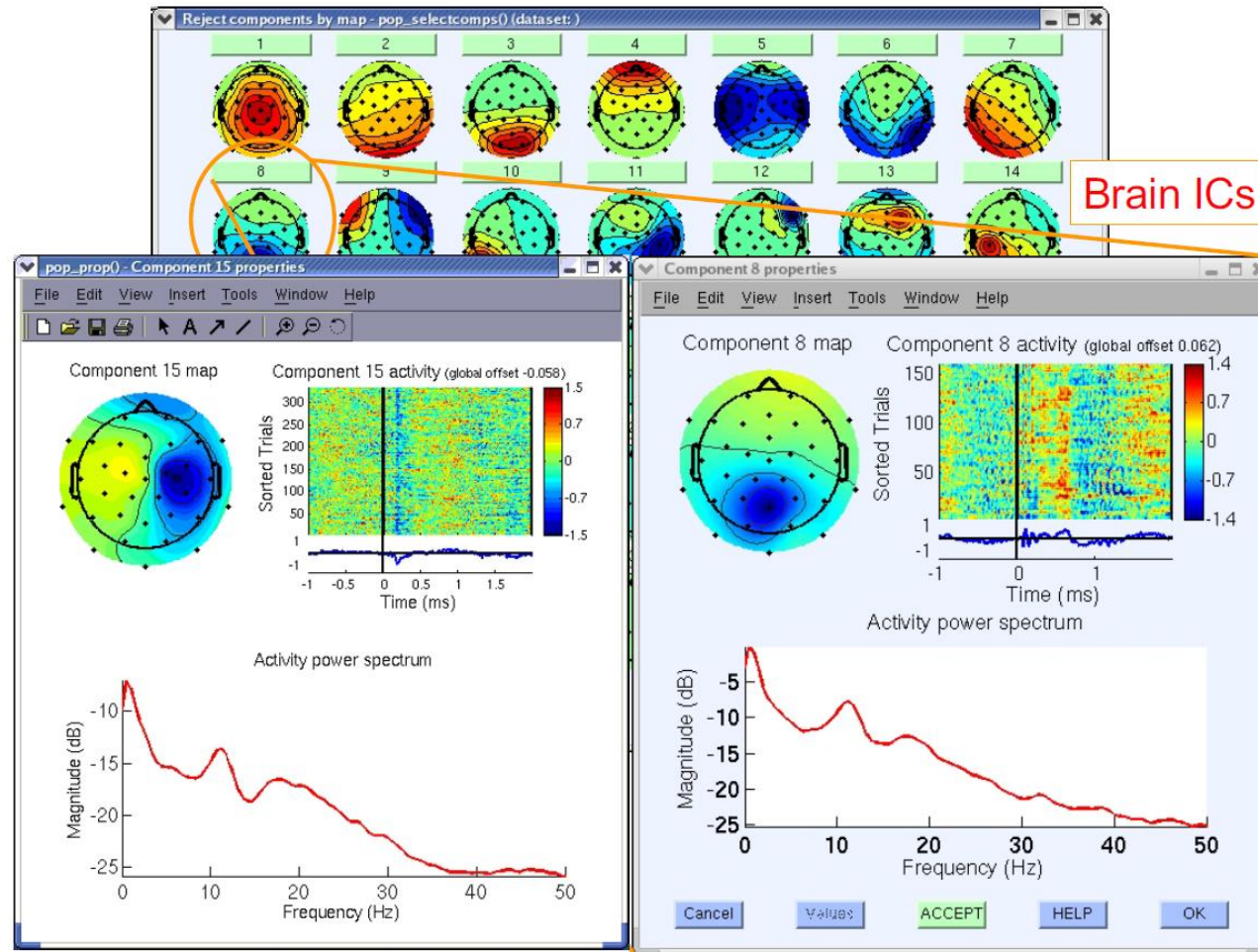


Examples of ICs – Pulse artifact

Pulse artifacts



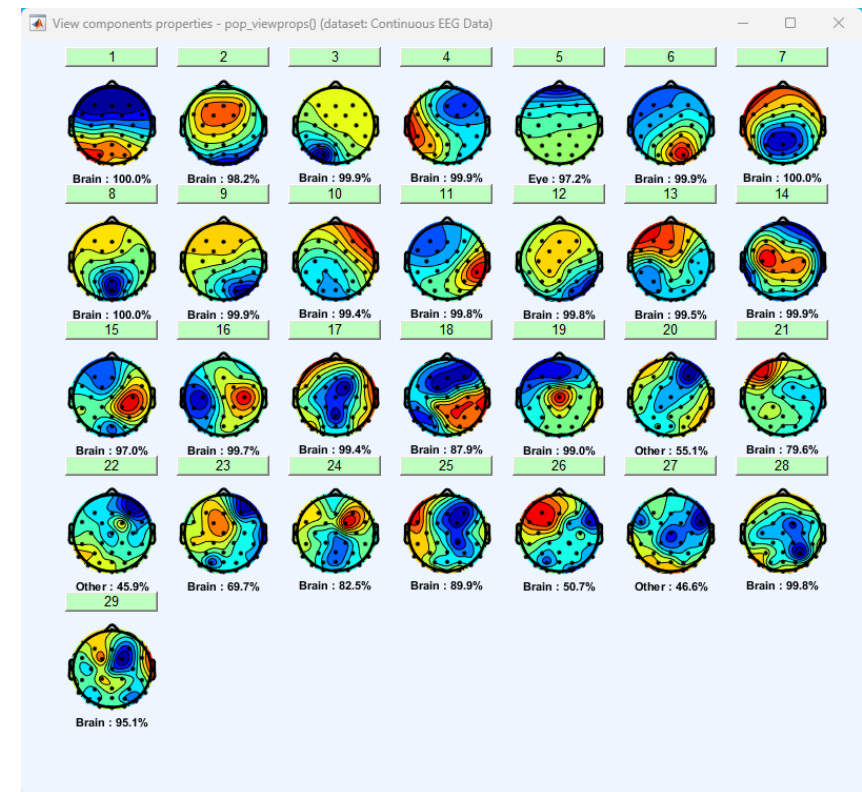
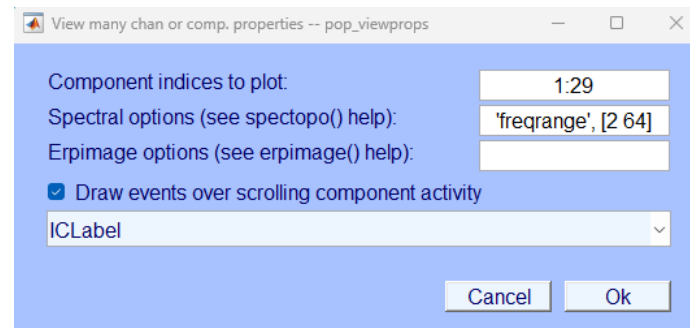
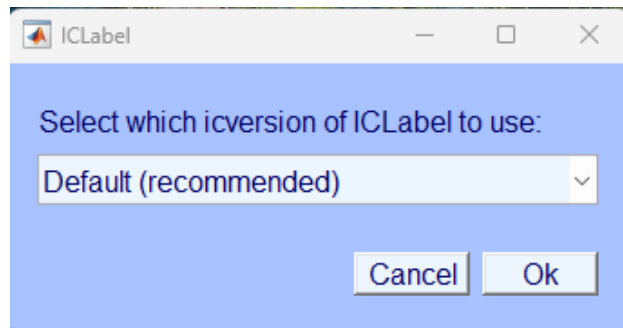
Examples of ICs – Brain signal



Label bad ICs automatically

Label components using

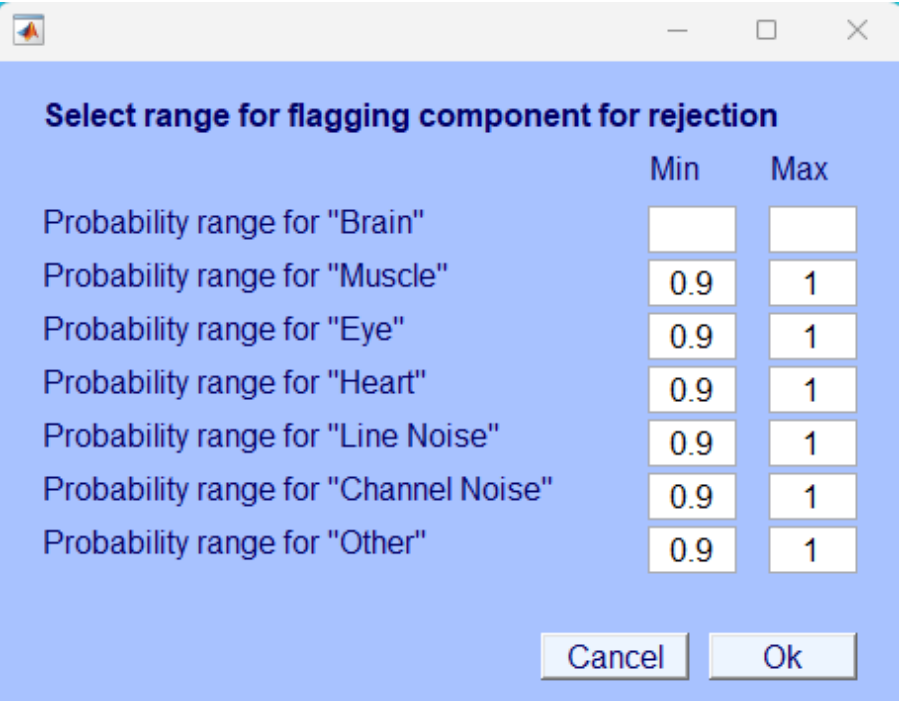
- Tools>Classify components using ICLabel>Label components



Classify bad ICs automatically

Classify components using

- **Tools>Classify components using IClabel>Flag components as artifact**



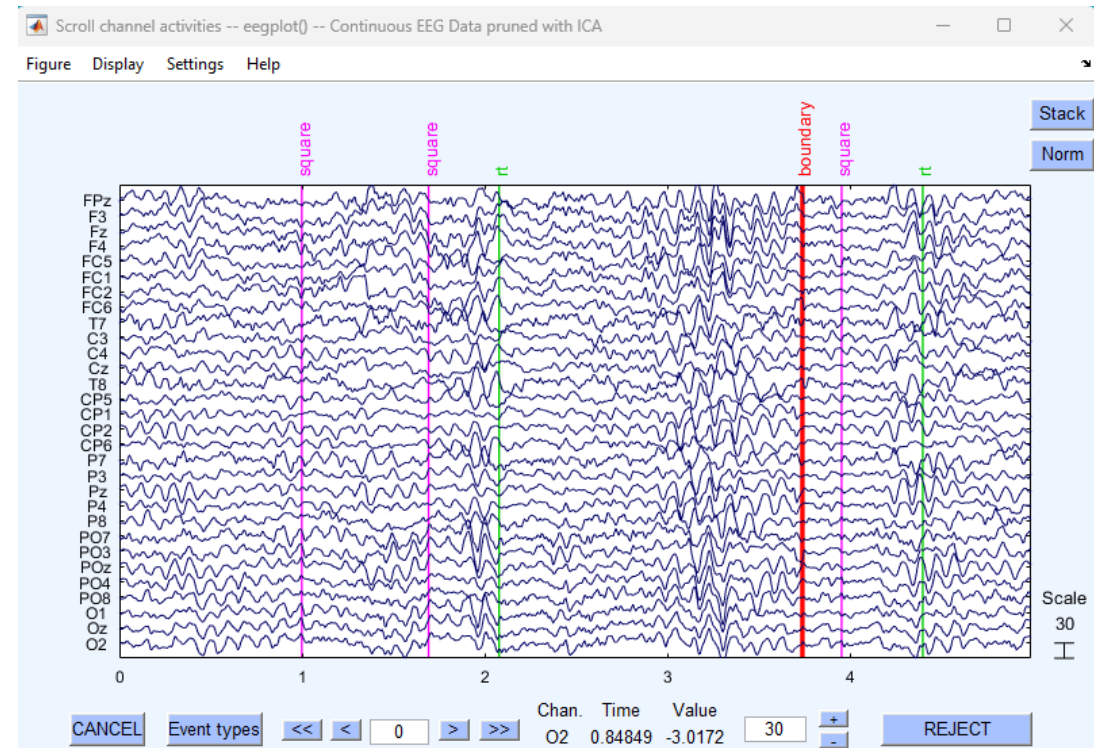
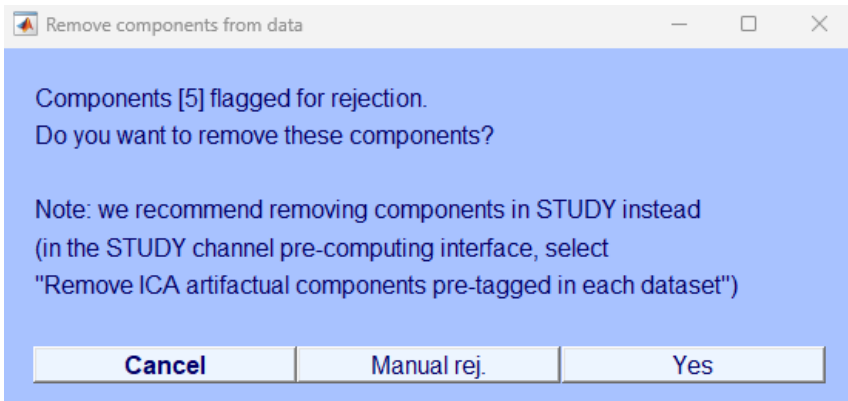
	Min	Max
Probability range for "Brain"		
Probability range for "Muscle"	0.9	1
Probability range for "Eye"	0.9	1
Probability range for "Heart"	0.9	1
Probability range for "Line Noise"	0.9	1
Probability range for "Channel Noise"	0.9	1
Probability range for "Other"	0.9	1

Cancel Ok

Reject bad ICs automatically

To reject bad ICs manually.

- **Tools>Remove components from data**



*Thank
You!*



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